

LoiTeS Development

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Obtaining and Formatting the Data

Here, we bring in the data from a csv copy of the raw xlsx file. This was done for anonymity to responses as IPAddress was originally captured. The data is formatted from wide to long for the purpose of factor analysis and correlation.

```
# raw xlsx had StartDate; EndDate; Status; IPAddress; Progress; Duration..in.seconds.; Finish
fn <- ('data/Loneliness Raw Data_May 20, 2026_12.39.csv')
d1 <- read_data(filename = fn, controlQ = 'Q30')
```

```
[1] "data/Loneliness Raw Data_May 20, 2026_12.39.csv"
[1] "Q30"
[1] "Participants after consent/age filter: 220"
[1] "Participants after complete/preview/age-99 filter: 218"
[1] "Inattentive total: 7"
[1] "IDs of inattentive participants:"
[1] NA NA NA NA NA NA NA
[1] "Number of accepted participants: 211"
```

```
instruments <- list(
  WHO_5 = c('Q28_1', 'Q28_2', 'Q28_3', 'Q28_4', 'Q28_5'),
  Michele = c('Q20_1', 'Q20_2', 'Q20_3', 'Q20_4', 'Q20_5',
    'Q20_6', 'Q20_7', 'Q20_8', 'Q20_9', 'Q20_10',
    'Q20_11', 'Q20_12'),
  Hughes = c('Q22_1', 'Q22_2', 'Q22_3', 'Q22_4', 'Q22_5',
    'Q22_6', 'Q22_7', 'Q22_8', 'Q22_9', 'Q22_10',
    'Q22_11', 'Q22_12', 'Q22_13', 'Q22_14', 'Q22_15',
    'Q22_16', 'Q22_17', 'Q22_18', 'Q22_19', 'Q22_20'),
```

```

Tschannen = c('Q24_1', 'Q24_2', 'Q24_3', 'Q24_4', 'Q24_5',
              'Q24_6', 'Q24_7', 'Q24_8', 'Q24_9', 'Q24_10',
              'Q24_11', 'Q24_12'),
Zimet      = c('Q26_1', 'Q26_2', 'Q26_3', 'Q26_4', 'Q26_5',
              'Q26_6', 'Q26_7', 'Q26_8', 'Q26_9', 'Q26_10',
              'Q26_11', 'Q26_12')
)

d1_long <- make_it_long(d1, instruments = instruments)

```

```

[1] "Number of rows in the long data: 12871"
[1] "Number of unique participants in the long data: 211"
[1] "Number of unique items in the long data: 61"
[1] "Instruments found: WHO_5, Michele, Hughes, Tschannen, Zimet"

```

Demographic Summary

There are some open questions regarding how to refer to the responses to the variables. Gender for example seems to be a binary choice (1,2) but it is unknown whether 1 = male or female. Similar issues are found for all of the responses to 'EmpCount', 'Role', 'Subject', 'LvlTaught', 'YearTaught', 'EmpStatus', 'TeachAct', 'OftenTeach', 'Qualified', 'AcadBack', 'Formal'. Once the response format is known the demographicV function can be written to handle all of these for the reporting.

Exploratory Factor Analysis

In this section, we explore the dimensionality of the Michele instrument. First, the data is cleaned to only contain the Michele instrument responses. Then KMO and Bartlett's tests are used to understand if the data is suitable for factor analysis. Second, the EFA is run for the cleaned data. Parallel analysis is used to understand how many factors are estimated to be in the data by simulating factor analyses. The number of factors for the actual EFA analysis is determined by this parallel simulation.

Prepping the data for EFA

Here, data is cleaned from the long format and to remove other variables than those in the Michele instrument. The KMO is good overall (.85) and for the individual items (range = .78-.92). The Bartlett's test ($\chi^2(66) = 1447.01, p < .001$) indicated items had sufficient correlations for factor analysis processes.

```
# We need temporarily remove the other instruments from the data frame because we only care a

d1_long_clean <- prepare_EFA(d1_long, instrument_name = "Michele")

cm <- cor(d1_long_clean, use = "pairwise.complete.obs", method = "pearson")
KMO(cm)
```

Kaiser-Meyer-Olkin factor adequacy

Call: KMO(r = cm)

Overall MSA = 0.85

MSA for each item =

	Q20_1	Q20_2	Q20_3	Q20_4	Q20_5	Q20_6	Q20_7	Q20_8	Q20_9	Q20_10	Q20_11
Q20_1	0.84	0.82	0.87	0.87	0.90	0.88	0.79	0.78	0.82	0.85	0.91
Q20_2		0.92									

```
cortest.bartlett(cm, n = nrow(d1_long_clean))
```

\$chisq

[1] 1447.011

\$p.value

[1] 7.71136e-259

\$df

[1] 66

Running an EFA

Before performing an EFA, parallel analysis was used to determine the number of factors to retain. This determined that a three factor solution best fit the data, following the simulations (shown on Scree Plots).

Factor analysis was then conducted on the 12 item Michele instrument using a Maximum Likelihood method (ML) with an oblimin rotation. Considering the sample size (> 200) ML was used to further investigate the statistical significance of the obtained factor loadings (Hair et al., 2019). An oblique rotation method was employed because it is assumed that factors will be correlated (Tabachnick & Fidell, 2019).

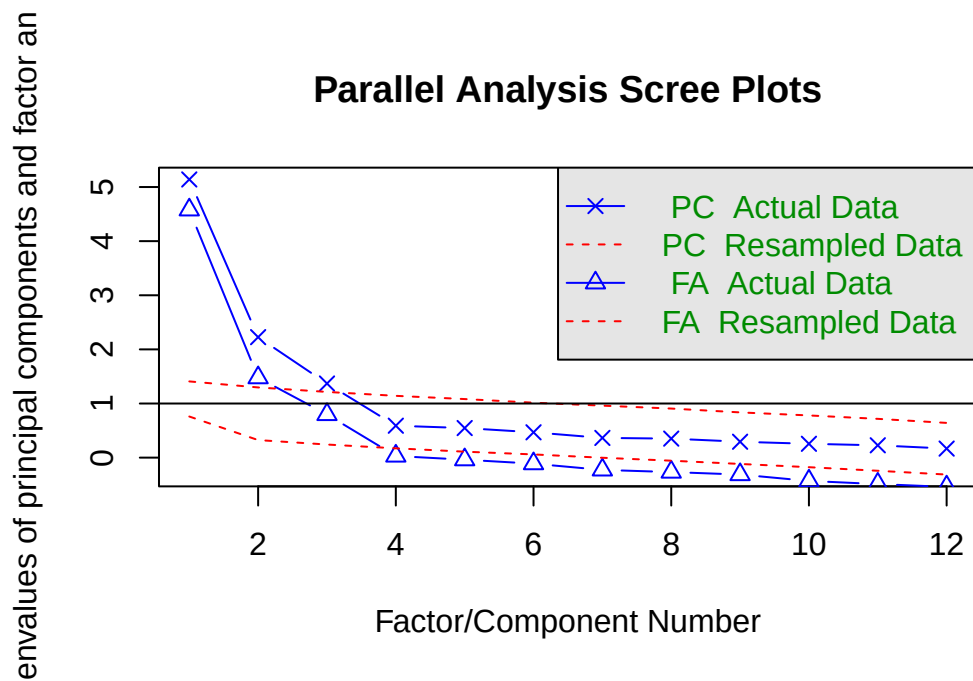
Together, the three factors explain 64% of the variance. Factor 1 explains 23% of the variance (4 items), as does Factor 2 (4 items). Factor 3 explains 18% of the total variance (4 items). The

communalities of the items, however, ranged from .389 (Q20_5) to .844 (Q20_10) ($M = .64$). Considering an item fell below the recommended threshold of .40 (Costello & Osborne, 2005), the item was removed before another factor analysis was performed. Additionally, another item (Q20_6)

```
set.seed(3108)

# obtaining communality descriptions
comm_table <- function(x) list(
  Mean = mean(x$communality),
  Median = median(x$communality),
  Range = range(x$communality))

# EFA simulation
output <- fa.parallel(d1_long_clean, fm = "ml", sim = FALSE)
```



Parallel analysis suggests that the number of factors = 3 and the number of components = 3.

```
# running EFA with the retained simulated factor structure (3 factors)
fa_sol_d1_long_clean <- fa(d1_long_clean, nfactors = output$nfact, n.iter = 0,
  rotate = "oblimin", missing = TRUE, warnings = TRUE,
  fm = "ml")
```

```
print(fa_sol_d1_long_clean)
```

Factor Analysis using method = ml

Call: fa(r = d1_long_clean, nfactors = output\$nfact, n.iter = 0, rotate = "oblimin",
missing = TRUE, warnings = TRUE, fm = "ml")

Standardized loadings (pattern matrix) based upon correlation matrix

	ML1	ML2	ML3	h2	u2	com
Q20_1	-0.07	0.83	0.12	0.69	0.31	1.1
Q20_2	-0.03	0.87	0.03	0.74	0.26	1.0
Q20_3	0.12	0.72	-0.01	0.60	0.40	1.1
Q20_4	0.10	0.81	-0.13	0.70	0.30	1.1
Q20_5	0.15	0.02	0.58	0.43	0.57	1.1
Q20_6	0.23	-0.10	0.52	0.39	0.61	1.5
Q20_7	0.00	0.00	0.83	0.68	0.32	1.0
Q20_8	-0.04	0.05	0.84	0.69	0.31	1.0
Q20_9	0.87	-0.08	0.06	0.74	0.26	1.0
Q20_10	0.92	-0.01	0.01	0.84	0.16	1.0
Q20_11	0.73	0.11	-0.07	0.58	0.42	1.1
Q20_12	0.66	0.20	0.05	0.62	0.38	1.2

	ML1	ML2	ML3
SS loadings	2.82	2.77	2.12
Proportion Var	0.23	0.23	0.18
Cumulative Var	0.23	0.47	0.64
Proportion Explained	0.37	0.36	0.27
Cumulative Proportion	0.37	0.73	1.00

With factor correlations of

	ML1	ML2	ML3
ML1	1.00	0.46	0.41
ML2	0.46	1.00	0.19
ML3	0.41	0.19	1.00

Mean item complexity = 1.1

Test of the hypothesis that 3 factors are sufficient.

df null model = 66 with the objective function = 7.05 with Chi Square = 1447.01
df of the model are 33 and the objective function was 0.3

The root mean square of the residuals (RMSR) is 0.02

The df corrected root mean square of the residuals is 0.03

The harmonic n.obs is 211 with the empirical chi square 17 with prob < 0.99
The total n.obs was 211 with Likelihood Chi Square = 61.33 with prob < 0.002

Tucker Lewis Index of factoring reliability = 0.959
RMSEA index = 0.064 and the 90 % confidence intervals are 0.038 0.089
BIC = -115.28
Fit based upon off diagonal values = 1
Measures of factor score adequacy

	ML1	ML2	ML3
Correlation of (regression) scores with factors	0.96	0.95	0.92
Multiple R square of scores with factors	0.92	0.90	0.85
Minimum correlation of possible factor scores	0.84	0.80	0.71

```
print(fa_sol_d1_long_clean$loadings, cutoff = 0.2)
```

Loadings:

	ML1	ML2	ML3
Q20_1		0.830	
Q20_2		0.873	
Q20_3		0.718	
Q20_4		0.807	
Q20_5			0.577
Q20_6	0.228		0.520
Q20_7			0.828
Q20_8			0.838
Q20_9	0.867		
Q20_10	0.916		
Q20_11	0.727		
Q20_12	0.655		

	ML1	ML2	ML3
SS loadings	2.654	2.686	2.034
Proportion Var	0.221	0.224	0.170
Cumulative Var	0.221	0.445	0.615

```
print(fa_sol_d1_long_clean$communality)
```

Q20_1	Q20_2	Q20_3	Q20_4	Q20_5	Q20_6	Q20_7	Q20_8
0.6862113	0.7449940	0.6030832	0.7036439	0.4314802	0.3899722	0.6837152	0.6930792

```

      Q20_9      Q20_10      Q20_11      Q20_12
0.7387192 0.8435024 0.5756296 0.6182806

```

```
comm_table(fa_sol_d1_long_clean)
```

```
$Mean
```

```
[1] 0.6426926
```

```
$Median
```

```
[1] 0.6849633
```

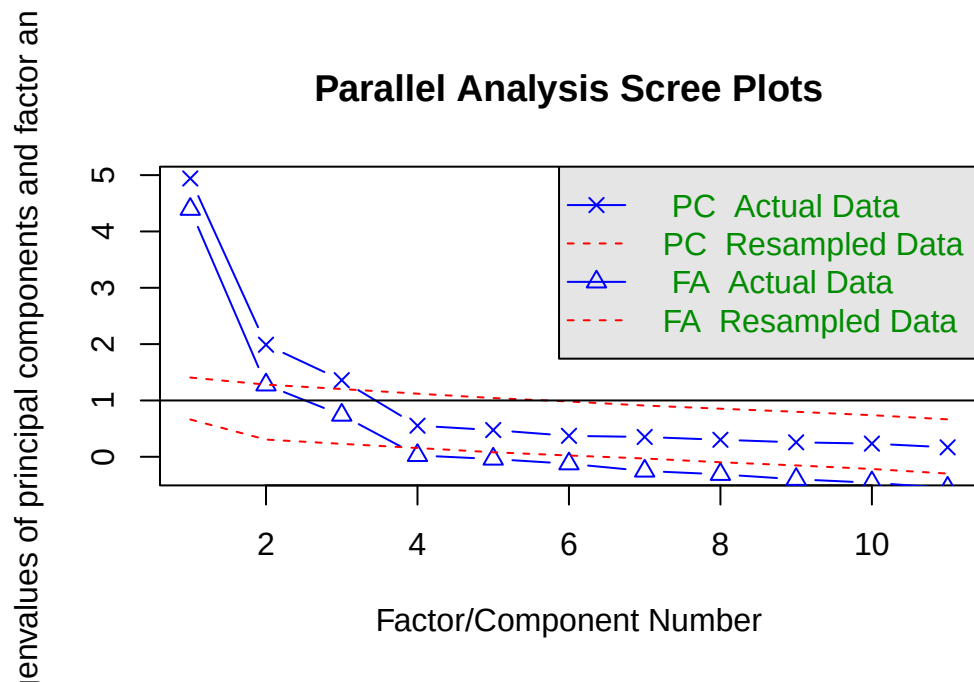
```
$Range
```

```
[1] 0.3899722 0.8435024
```

```
# Q20_5 has poor communality which hinders the factor loading. Q20_6 had a cross-loading with
```

```
d1_long_clean2 <- dplyr::select(d1_long_clean, -Q20_6)
```

```
output <- fa.parallel(d1_long_clean2, fm = "ml",sim = FALSE)
```



```
Parallel analysis suggests that the number of factors = 3 and the number of components = 3
```

```
fa_sol_d1_long_clean2 <- fa(d1_long_clean2, nfactors = output$nfact, n.iter = 0,
                             rotate = "oblimin", missing = TRUE, warnings = TRUE,
                             fm = "ml")

print(fa_sol_d1_long_clean2)
```

Factor Analysis using method = ml

Call: fa(r = d1_long_clean2, nfactors = output\$nfact, n.iter = 0, rotate = "oblimin",
missing = TRUE, warnings = TRUE, fm = "ml")

Standardized loadings (pattern matrix) based upon correlation matrix

	ML2	ML1	ML3	h2	u2	com
Q20_1	0.83	-0.07	0.11	0.69	0.31	1.1
Q20_2	0.87	-0.04	0.03	0.74	0.26	1.0
Q20_3	0.72	0.12	-0.02	0.61	0.39	1.1
Q20_4	0.81	0.09	-0.12	0.70	0.30	1.1
Q20_5	0.01	0.17	0.55	0.41	0.59	1.2
Q20_7	-0.01	0.04	0.79	0.64	0.36	1.0
Q20_8	0.03	-0.03	0.88	0.76	0.24	1.0
Q20_9	-0.09	0.87	0.06	0.74	0.26	1.0
Q20_10	-0.01	0.92	0.01	0.84	0.16	1.0
Q20_11	0.11	0.73	-0.08	0.58	0.42	1.1
Q20_12	0.19	0.66	0.05	0.62	0.38	1.2

	ML2	ML1	ML3
SS loadings	2.78	2.76	1.79
Proportion Var	0.25	0.25	0.16
Cumulative Var	0.25	0.50	0.67
Proportion Explained	0.38	0.38	0.24
Cumulative Proportion	0.38	0.76	1.00

With factor correlations of

	ML2	ML1	ML3
ML2	1.00	0.47	0.20
ML1	0.47	1.00	0.39
ML3	0.20	0.39	1.00

Mean item complexity = 1.1

Test of the hypothesis that 3 factors are sufficient.

df null model = 55 with the objective function = 6.58 with Chi Square = 1351.5
df of the model are 25 and the objective function was 0.24

The root mean square of the residuals (RMSR) is 0.02
The df corrected root mean square of the residuals is 0.03

The harmonic n.obs is 211 with the empirical chi square 11.42 with prob < 0.99
The total n.obs was 211 with Likelihood Chi Square = 48.37 with prob < 0.0034

Tucker Lewis Index of factoring reliability = 0.96
RMSEA index = 0.066 and the 90 % confidence intervals are 0.038 0.095
BIC = -85.43

Fit based upon off diagonal values = 1

Measures of factor score adequacy

	ML2	ML1	ML3
Correlation of (regression) scores with factors	0.95	0.96	0.92
Multiple R square of scores with factors	0.90	0.92	0.85
Minimum correlation of possible factor scores	0.80	0.84	0.70

```
print(fa_sol_d1_long_clean2$loadings, cutoff = 0.2)
```

Loadings:

	ML2	ML1	ML3
Q20_1	0.833		
Q20_2	0.873		
Q20_3	0.722		
Q20_4	0.807		
Q20_5			0.550
Q20_7			0.789
Q20_8			0.876
Q20_9		0.872	
Q20_10		0.918	
Q20_11		0.731	
Q20_12		0.659	

	ML2	ML1	ML3
SS loadings	2.686	2.632	1.734
Proportion Var	0.244	0.239	0.158
Cumulative Var	0.244	0.483	0.641

```
print(fa_sol_d1_long_clean2$communality)
```

Q20_1	Q20_2	Q20_3	Q20_4	Q20_5	Q20_7	Q20_8	Q20_9
-------	-------	-------	-------	-------	-------	-------	-------

```
0.6873241 0.7436474 0.6050468 0.6980912 0.4097655 0.6424319 0.7599348 0.7410055
  Q20_10    Q20_11    Q20_12
0.8426832 0.5778698 0.6179999
```

```
comm_table(fa_sol_d1_long_clean2)
```

```
$Mean
```

```
[1] 0.6659818
```

```
$Median
```

```
[1] 0.6873241
```

```
$Range
```

```
[1] 0.4097655 0.8426832
```

```
psych::alpha(d1_long_clean2)
```

```
Reliability analysis
```

```
Call: psych::alpha(x = d1_long_clean2)
```

raw_alpha	std.alpha	G6(smc)	average_r	S/N	ase	mean	sd	median_r
0.88	0.87	0.92	0.38	6.9	0.012	3.7	0.66	0.33

```
95% confidence boundaries
```

```
lower alpha upper
```

```
Feldt      0.85 0.88 0.9
```

```
Duhachek   0.85 0.88 0.9
```

```
Reliability if an item is dropped:
```

	raw_alpha	std.alpha	G6(smc)	average_r	S/N	alpha	se	var.r	med.r
Q20_1	0.86	0.86	0.90	0.38	6.2	0.013	0.037	0.33	
Q20_2	0.86	0.86	0.90	0.38	6.2	0.013	0.035	0.32	
Q20_3	0.86	0.86	0.91	0.38	6.1	0.014	0.037	0.32	
Q20_4	0.86	0.86	0.91	0.38	6.3	0.013	0.033	0.32	
Q20_5	0.88	0.87	0.92	0.40	6.7	0.012	0.039	0.33	
Q20_7	0.88	0.87	0.91	0.40	6.8	0.012	0.036	0.37	
Q20_8	0.88	0.87	0.91	0.40	6.8	0.012	0.036	0.37	
Q20_9	0.86	0.86	0.90	0.38	6.0	0.014	0.037	0.33	
Q20_10	0.86	0.85	0.90	0.37	5.8	0.014	0.037	0.31	
Q20_11	0.86	0.86	0.91	0.38	6.1	0.014	0.038	0.32	

Q20_12	0.86	0.85	0.90	0.37	5.8	0.014	0.039	0.32
--------	------	------	------	------	-----	-------	-------	------

Item statistics

	n	raw.r	std.r	r.cor	r.drop	mean	sd
Q20_1	211	0.69	0.68	0.65	0.60	3.5	1.08
Q20_2	211	0.70	0.68	0.66	0.62	3.7	0.99
Q20_3	211	0.70	0.69	0.66	0.62	3.7	1.00
Q20_4	211	0.69	0.66	0.64	0.60	3.6	1.07
Q20_5	211	0.50	0.55	0.49	0.42	4.1	0.69
Q20_7	211	0.47	0.54	0.50	0.40	4.2	0.69
Q20_8	211	0.48	0.55	0.51	0.40	4.3	0.71
Q20_9	211	0.73	0.71	0.70	0.64	3.5	1.14
Q20_10	211	0.79	0.77	0.77	0.72	3.6	1.09
Q20_11	211	0.73	0.70	0.66	0.64	3.2	1.16
Q20_12	211	0.79	0.77	0.74	0.72	3.4	1.08

Non missing response frequency for each item

	1	2	3	4	5	miss
Q20_1	0.06	0.12	0.21	0.46	0.15	0
Q20_2	0.04	0.09	0.15	0.55	0.17	0
Q20_3	0.04	0.11	0.14	0.55	0.16	0
Q20_4	0.04	0.15	0.16	0.48	0.18	0
Q20_5	0.00	0.02	0.09	0.60	0.29	0
Q20_7	0.00	0.02	0.09	0.60	0.29	0
Q20_8	0.00	0.03	0.05	0.46	0.45	0
Q20_9	0.06	0.15	0.15	0.45	0.18	0
Q20_10	0.04	0.17	0.14	0.47	0.19	0
Q20_11	0.10	0.19	0.25	0.35	0.11	0
Q20_12	0.08	0.12	0.26	0.43	0.12	0

There are three factors with strong items in each factor. While removing the two low commun

Correlations between instruments

Requires: d1 (wide format, as returned by read_data())

```
# make sure everything is numeric!
d1_cor <- d1 %>% mutate(across(matches("^Q(20|22|24|26|28)_"), as.numeric))

# Need to calculate the scores for the instruments prior to correlation
```

```

# Dealing with reverse-scoring items

# Hughes (R-UCLA): items 1,4,5,6,9,10,15,16,19,20 are reverse-scored.
hughes_rev <- c('Q22_1', 'Q22_4', 'Q22_5', 'Q22_6',
               'Q22_9', 'Q22_10', 'Q22_15', 'Q22_16',
               'Q22_19', 'Q22_20')

d1_cor <- d1_cor %>%
  mutate(across(all_of(hughes_rev), ~ 5 - as.numeric(.)))

# Compute scale / sub-scale scores
# (row means → comparable across instruments of different lengths)

score_df <- d1_cor %>%
  mutate(

    # WHO-5 (total score: raw 0-25, × 4 = 0-100)
    WHO5_total = rowSums(
      across(c('Q28_1', 'Q28_2', 'Q28_3', 'Q28_4', 'Q28_5')),
      na.rm = TRUE) * 4,

    # Michele (full scale + EFA-derived sub-scales)
    Michele_full = rowMeans(
      across(c('Q20_1', 'Q20_2', 'Q20_3', 'Q20_4',
               'Q20_5', 'Q20_6', 'Q20_7', 'Q20_8',
               'Q20_9', 'Q20_10', 'Q20_11', 'Q20_12'))),
      na.rm = TRUE),

    # EFA Factor 1
    Michele_F1 = rowMeans(
      across(c('Q20_1', 'Q20_2', 'Q20_3', 'Q20_4')),
      na.rm = TRUE),

    # EFA Factor 2
    Michele_F2 = rowMeans(
      across(c('Q20_9', 'Q20_10', 'Q20_11', 'Q20_12')),
      na.rm = TRUE),

    # EFA Factor 3
    Michele_F3 = rowMeans(
      across(c('Q20_5', 'Q20_7', 'Q20_8')),

```

```

    na.rm = TRUE),

# Hughes / R-UCLA (sum score after reverse-scoring)
Hughes_total = rowSums(
  across(c('Q22_1', 'Q22_2', 'Q22_3', 'Q22_4', 'Q22_5',
            'Q22_6', 'Q22_7', 'Q22_8', 'Q22_9', 'Q22_10',
            'Q22_11', 'Q22_12', 'Q22_13', 'Q22_14', 'Q22_15',
            'Q22_16', 'Q22_17', 'Q22_18', 'Q22_19', 'Q22_20')),
  na.rm = TRUE),

# Tschannen-Moran (3 sub-scales, mean score each)
Tschannen_total = rowMeans(
  across(c('Q24_1', 'Q24_2', 'Q24_3', 'Q24_4', 'Q24_5',
            'Q24_6', 'Q24_7', 'Q24_8', 'Q24_9', 'Q24_10',
            'Q24_11', 'Q24_12')),
  na.rm = TRUE),

# Sub-scale 1
Tschannen_instruction = rowMeans(
  across(c('Q24_5', 'Q24_9', 'Q24_10', 'Q24_12')),
  na.rm = TRUE),

# Sub-scale 2
Tschannen_class_manager = rowMeans(
  across(c('Q24_1', 'Q24_3', 'Q24_6', 'Q24_8')),
  na.rm = TRUE),

# Sub-scale 3
Tschannen_engagement = rowMeans(
  across(c('Q24_2', 'Q24_4', 'Q24_7', 'Q24_11')),
  na.rm = TRUE),

# Zimet MSPSS (3 sub-scales, mean score / 4)
Zimet_total = rowMeans(
  across(c('Q26_1', 'Q26_2', 'Q26_3', 'Q26_4', 'Q26_5',
            'Q26_6', 'Q26_7', 'Q26_8', 'Q26_9', 'Q26_10',
            'Q26_11', 'Q26_12')),
  na.rm = TRUE),

# Sub-scale 1
Zimet_sig_other = rowMeans(
  across(c('Q26_1', 'Q26_2', 'Q26_5', 'Q26_10')),

```

```

    na.rm = TRUE),

# Sub-scale 2
Zimet_family    = rowMeans(
  across(c('Q26_3', 'Q26_4', 'Q26_8', 'Q26_11')),
  na.rm = TRUE),

# Sub-scale 3
Zimet_friends   = rowMeans(
  across(c('Q26_6', 'Q26_7', 'Q26_9', 'Q26_12')),
  na.rm = TRUE)
)

# Pull score columns into a correlation matrix

score_vars <- c(
  "WHO5_total",
  "Michele_full",  "Michele_F1",  "Michele_F2",  "Michele_F3",
  "Hughes_total",
  "Tschannen_total", "Tschannen_instruction", "Tschannen_class_manager", "Tschannen_engagement",
  "Zimet_total",    "Zimet_sig_other", "Zimet_family", "Zimet_friends"
)

cor_data <- score_df %>%
  dplyr::select(all_of(score_vars))

# Pearson correlation matrix with pairwise complete observations
cor_matrix <- cor(cor_data, use = "pairwise.complete.obs", method = "pearson")

# p-values via psych::corr.test (also gives CIs)
cor_test <- psych::corr.test(cor_data, use = "pairwise", method = "pearson",
                             adjust = "holm") # Holm correction for multiple comparisons

# Display: correlation table with significance stars

# attach stars to a correlation matrix
add_stars <- function(r_mat, p_mat) {
  stars <- ifelse(p_mat < .001, "***",
    ifelse(p_mat < .010, "**",
    ifelse(p_mat < .050, "*", "")))
  matrix(

```

```

    paste0(sprintf("%.2f", r_mat), stars),
    nrow = nrow(r_mat),
    dimnames = dimnames(r_mat)
  )
}

cor_starred <- add_stars(cor_matrix, cor_test$p)
cor_starred[upper.tri(cor_starred, diag = TRUE)] <- ""

# Short display labels (edit to match your preferred names)
rownames(cor_starred) <- colnames(cor_starred) <- c(
  "WHO-5",
  "LoiTeS", "LoiTeS F1", "LoiTeS F2", "LoiTeS F3",
  "Hughes",
  "Tschannen", "Instruction", "Cl. Mgmt", "Engagement",
  "Zimet", "Sig. Other", "Family", "Friends"
)

kableExtra::kbl(
  cor_starred,
  caption = "Pearson correlations between scale and sub-scale scores",
  booktabs = TRUE,
  align = "r"
) %>%
  kableExtra::kable_styling(
    latex_options = c("scale_down", "hold_position"),
    font_size = 9
  ) %>%
  kableExtra::footnote(
    general = "* p < .05 ** p < .01 *** p < .001 (Holm-corrected). Diagonals omitted.",
    general_title = "Note.",
    footnote_as_chunk = TRUE
  )

# Descriptive statistics for each score

desc_stats <- cor_data %>%
  tidyr::pivot_longer(everything(), names_to = "Scale", values_to = "score") %>%
  dplyr::group_by(Scale) %>%
  dplyr::summarise(
    N = sum(!is.na(score)),
    Mean = round(mean(score, na.rm = TRUE), 2),

```

```

    SD    = round(sd(score,   na.rm = TRUE), 2),
    Min   = round(min(score,  na.rm = TRUE), 2),
    Max   = round(max(score,  na.rm = TRUE), 2),
    .groups = "drop"
  ) %>%
  dplyr::arrange(match(Scale, score_vars)) # keep original order

kableExtra::kbl(
  desc_stats,
  caption = "Descriptive statistics for scale and sub-scale scores",
  booktabs = TRUE,
  digits    = 2
) %>%
  kableExtra::kable_styling(latex_options = "hold_position")

```

Linear Mixed Models

Want to compare how the variables for the teacher demographics relate to the responses to the WHO-5, LoiTeS, Hughes, Tschannen, and Zimet instruments.

This section is not correct. Need to get the renamed variable groups as the comparisons rather than the raw. The issue is there are multi-select responses which make no sense to have individual...

```

extract_fixed <- function(model, outcome_name) {
  coef_table <- coef(summary(model))
  ci          <- confint(model, parm = "beta_", method = "Wald")
  data.frame(
    outcome   = outcome_name,
    term      = rownames(coef_table),
    estimate  = coef_table[, "Estimate"],
    std.error = coef_table[, "Std. Error"],
    statistic = coef_table[, "t value"],
    p.value   = coef_table[, "Pr(>|t|)"],
    conf.low  = ci[rownames(coef_table), 1],
    conf.high = ci[rownames(coef_table), 2],
    row.names = NULL
  )
}

```

```

outcomes <- c(
  "WHO5_total",
  "Michele_full", "Michele_F1", "Michele_F2", "Michele_F3",
  "Hughes_total",
  "Tschannen_total", "Tschannen_instruction",
  "Tschannen_class_manager", "Tschannen_engagement",
  "Zimet_total", "Zimet_sig_other", "Zimet_family", "Zimet_friends"
)

demo_vars <- c("Age", "Gender", "Role", "LvlTaught", "YearTaught",
  "EmpStatus", "TeachAct", "OftenTeach",
  "Qualified", "Formal")

# Fit one model per outcome -----
broad_models <- lapply(setNames(outcomes, outcomes), function(outcome) {
  fml <- as.formula(
    paste(outcome, "~", paste(demo_vars, collapse = " + "),
      "+ (1 | EmpCount)")
  )
  lmer(fml, data = score_df, REML = FALSE,
    control = lmerControl(optimizer = "bobyqa"))
})

```

fixed-effect model matrix is rank deficient so dropping 2 columns / coefficients
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boundary (singular) fit: see help('isSingular')

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fixed-effect model matrix is rank deficient so dropping 2 columns / coefficients
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```
# Extract fixed effects (no broom.mixed) -----  
broad_tidy <- bind_rows(  
  mapapply(extract_fixed,  
    model      = broad_models,  
    outcome_name = names(broad_models),  
    SIMPLIFY    = FALSE)  
)  
  
# Significant predictors at p < .05 (Satterthwaite)
```

```

sig_predictors <- broad_tidy %>%
  filter(p.value < .05, term != "(Intercept)") %>%
  dplyr::select(outcome, term, estimate, std.error, statistic, p.value) %>%
  arrange(outcome, p.value)

print(sig_predictors)

```

	outcome
1	Hughes_total
2	Hughes_total
3	Hughes_total
4	Hughes_total
5	Hughes_total
6	Hughes_total
7	Hughes_total
8	Hughes_total
9	Hughes_total
10	Hughes_total
11	Hughes_total
12	Hughes_total
13	Hughes_total
14	Hughes_total
15	Hughes_total
16	Hughes_total
17	Hughes_total
18	Hughes_total
19	Hughes_total
20	Hughes_total
21	Hughes_total
22	Hughes_total
23	Michele_F1
24	Michele_F1
25	Michele_F1
26	Michele_F1
27	Michele_F1
28	Michele_F2
29	Michele_F2
30	Michele_F2
31	Michele_F2
32	Michele_F3
33	Michele_F3
34	Michele_F3

35	Michele_F3
36	Michele_F3
37	Michele_F3
38	Michele_F3
39	Michele_F3
40	Michele_F3
41	Michele_F3
42	Michele_F3
43	Michele_F3
44	Michele_F3
45	Michele_F3
46	Michele_F3
47	Michele_F3
48	Michele_full
49	Michele_full
50	Michele_full
51	Michele_full
52	Michele_full
53	Michele_full
54	Michele_full
55	Tschannen_class_manager
56	Tschannen_class_manager
57	Tschannen_class_manager
58	Tschannen_class_manager
59	Tschannen_class_manager
60	Tschannen_class_manager
61	Tschannen_engagement
62	Tschannen_engagement
63	Tschannen_engagement
64	Tschannen_engagement
65	Tschannen_engagement
66	Tschannen_engagement
67	Tschannen_engagement
68	Tschannen_engagement
69	Tschannen_instruction
70	Tschannen_instruction
71	Tschannen_instruction
72	Tschannen_instruction
73	Tschannen_instruction
74	Tschannen_instruction
75	Tschannen_total
76	Tschannen_total
77	Tschannen_total

78	Tschannen_total
79	Tschannen_total
80	Tschannen_total
81	Tschannen_total
82	Tschannen_total
83	Tschannen_total
84	Tschannen_total
85	WH05_total
86	WH05_total
87	WH05_total
88	WH05_total
89	WH05_total
90	WH05_total
91	WH05_total
92	WH05_total
93	WH05_total
94	WH05_total
95	WH05_total
96	WH05_total
97	WH05_total
98	WH05_total
99	WH05_total
100	Zimet_family
101	Zimet_family
102	Zimet_family
103	Zimet_family
104	Zimet_family
105	Zimet_family
106	Zimet_friends
107	Zimet_friends
108	Zimet_friends
109	Zimet_friends
110	Zimet_friends
111	Zimet_friends
112	Zimet_friends
113	Zimet_friends
114	Zimet_friends
115	Zimet_friends
116	Zimet_friends
117	Zimet_friends
118	Zimet_friends
119	Zimet_friends
120	Zimet_friends

121	Zimet_friends
122	Zimet_friends
123	Zimet_friends
124	Zimet_friends
125	Zimet_sig_other
126	Zimet_sig_other
127	Zimet_sig_other
128	Zimet_sig_other
129	Zimet_sig_other
130	Zimet_sig_other
131	Zimet_sig_other
132	Zimet_sig_other
133	Zimet_sig_other
134	Zimet_sig_other
135	Zimet_sig_other
136	Zimet_sig_other
137	Zimet_sig_other
138	Zimet_sig_other
139	Zimet_total
140	Zimet_total
141	Zimet_total
142	Zimet_total
143	Zimet_total
144	Zimet_total
145	Zimet_total
146	Zimet_total
147	Zimet_total
148	Zimet_total
149	Zimet_total

1	TeachActOne-to-one
2	EmpStatus
3	LvlTaughtEarly Childhood
4	LvlTaught
5	EmpStatus
6	TeachActClassroom-based / Small group / Pract
7	TeachActClassroom-based
8	LvlTaughtEarly Childhood / Primary / Lower secondary / U
9	RoleClassroom teacher / Tea
10	LvlTaughtLower secondary / U
11	EmpStatusFreelance
12	
13	RoleSubject Specialist / Teaching Ass

57	LvlTaughtEarly Childhood / Primary / Lower secondary / Upper secondary / Full-time
58	RoleSubject Specialist / Special Education
59	LvlTaughtLower secondary / Upper secondary / Full-time
60	RoleClassroom teacher / Teacher
61	TeachActOne-to-one / Online/Blended / Practitioner
62	RoleClassroom teacher / Teacher
63	TeachActOne-to-one / Online/Blended / Practitioner
64	LvlTaughtLower secondary / Upper secondary / Full-time
65	RoleSubject Specialist / Special Education
66	LvlTaughtLower secondary / Upper secondary / Full-time
67	TeachActClassroom-based / One-to-one / Online/Blended / Practitioner
68	RoleClassroom teacher / Teacher
69	TeachActClassroom-based / One-to-one / Online/Blended / Practitioner
70	TeachActClassroom-based / One-to-one / Online/Blended / Practitioner
71	RoleClassroom teacher / Teacher
72	EmpStatusEmployed / Unemployed
73	TeachActClassroom-based / One-to-one / Online/Blended / Practitioner
74	TeachActClassroom-based / One-to-one / Online/Blended / Practitioner
75	TeachActClassroom-based / One-to-one / Online/Blended / Practitioner
76	RoleClassroom teacher / Teacher
77	RoleSubject Specialist / Special Education
78	RoleClassroom teacher / Teacher
79	RoleSubject Specialist / Special Education
80	TeachActClassroom-based / One-to-one / Online/Blended / Practitioner
81	TeachActClassroom-based / One-to-one / Online/Blended / Practitioner
82	LvlTaughtEarly Childhood / Primary / Lower secondary / Upper secondary / Full-time
83	TeachActSmall group / One-to-one / Online/Blended / Practitioner
84	TeachActOne-to-one / Online/Blended / Practitioner
85	RoleClassroom teacher / Teacher
86	RoleSubject Specialist / Special Education
87	RoleSubject Specialist / Special Education
88	EmpStatusEmployed / Unemployed
89	TeachActClassroom-based / One-to-one / Online/Blended / Practitioner
90	TeachActClassroom-based / One-to-one / Online/Blended / Practitioner
91	YearTaughtYear 1 / Year 2 / Year 3 / Year 4 / Year 5 / Year 6 / Year 7 / Year 8 / Year 9 / Year 10 / Year 11 / Year 12
92	RoleSubject Specialist / Teaching Assistant
93	EmpStatusEmployed / Unemployed
94	TeachActClassroom-based / One-to-one / Online/Blended / Practitioner
95	RoleSubject Specialist / Special Education
96	TeachActSmall group / One-to-one / Online/Blended / Practitioner
97	TeachActClassroom-based / One-to-one / Online/Blended / Practitioner
98	TeachActClassroom-based / One-to-one / Online/Blended / Practitioner
99	TeachActOne-to-one / Online/Blended / Practitioner

100	TeachActOne-to-one / Small group /
101	TeachActClassroom-based / One-to-one /
102	TeachActClassroom-based
103	TeachActOne-to-one /
104	TeachActClassroom-based / One-to-one / Small group / Prac
105	LvlTaughtEarly Child
106	EmpStatus
107	EmpStatus
108	EmpStatusFreelance
109	RoleSubject Spec
110	LvlTaughtFun
111	TeachActClassroom-based
112	
113	LvlTaught
114	
115	
116	LvlTaughtPrimary / Lower secondary / Upper se
117	LvlTaught
118	QualifiedProfessional
119	LvlTaughtPrimary / Lower secondary / Upper secondary / Fun
120	LvlTaughtEarly Childhood / Primary /
121	
122	LvlTaughtEarly Child
123	TeachActOne-to-one /
124	LvlTaughtPrimary / Upper secondary / Fun
125	TeachActClassroom-based
126	RoleSupport/Special / Principal/Coor
127	OftenTeachIrregular/var
128	TeachAct
129	
130	EmpStatus
131	LvlTaughtEarly Child
132	RoleSubject Specialist / Support/Special / Tea
133	YearTaught
134	EmpStatus
135	LvlTaught
136	LvlTaughtEarly Childhood / Primary / Lower secondary / U
137	Role
138	LvlTaught
139	TeachActClassroom-based
140	
141	TeachActOne-to-one /
142	LvlTaughtEarly Child

143 EmpStatusL
 144 LvlTaughtL
 145 EmpStatusL
 146 EmpStatusFreelance
 147 LvlTaughtEarly Childhood / Primary / Lower secondary / U
 148 RoleSubject Specialist / Support/Special / Teac
 149 TeachActClassroom-based

	estimate	std.error	statistic	p.value
1	36.32835487	8.895749784	4.083788	6.297553e-05
2	-15.90013983	4.670192134	-3.404601	8.046512e-04
3	-18.03001642	5.302284600	-3.400424	8.065410e-04
4	-34.34470829	10.107489388	-3.397947	8.126373e-04
5	-13.54666409	4.219332459	-3.210618	1.560832e-03
6	13.65992140	4.420989804	3.089788	2.273690e-03
7	12.90832793	4.290350961	3.008688	2.947716e-03
8	-18.51104685	6.560129039	-2.821750	5.234304e-03
9	-28.23765656	10.209319139	-2.765871	6.210977e-03
10	-13.66390931	5.018478833	-2.722719	7.016548e-03
11	-15.74590564	5.972050916	-2.636599	9.000794e-03
12	-19.92026348	7.754428349	-2.568889	1.091565e-02
13	19.68648048	8.034291198	2.450307	1.516762e-02
14	-19.05123546	7.829167007	-2.433367	1.581123e-02
15	-21.34874148	8.783505663	-2.430549	1.597312e-02
16	-14.97676167	6.323937275	-2.368265	1.877494e-02
17	-3.46924371	1.568576360	-2.211715	2.859774e-02
18	19.96003479	9.399865259	2.123438	3.491811e-02
19	-18.75097189	8.894469354	-2.108161	3.625536e-02
20	4.58484054	2.202776911	2.081391	3.861355e-02
21	-8.49529804	4.140685316	-2.051665	4.144097e-02
22	-5.32694033	2.682287777	-1.985969	4.833142e-02
23	-2.86257666	0.905189858	-3.162405	1.795185e-03
24	-2.21730642	0.781806149	-2.836133	5.010913e-03
25	-2.60695957	0.943247993	-2.763811	6.218304e-03
26	-2.55058780	1.093020384	-2.333523	2.056186e-02
27	-0.55285241	0.238413220	-2.318883	2.135848e-02
28	-2.23852089	0.825232436	-2.712594	7.236070e-03
29	-0.90856119	0.354322384	-2.564222	1.108512e-02
30	-3.33385189	1.356040502	-2.458519	1.475977e-02
31	-0.55843677	0.249200512	-2.240913	2.610436e-02
32	0.92687855	0.277004468	3.346078	9.702033e-04
33	-0.01992034	0.006797583	-2.930503	3.756694e-03
34	1.29536772	0.442849324	2.925076	3.820205e-03
35	1.61245066	0.559795139	2.880430	4.381078e-03

36	1.70872601	0.595313274	2.870297	4.518415e-03
37	1.50821864	0.526582705	2.864163	4.603456e-03
38	0.93693720	0.372367875	2.516160	1.260863e-02
39	1.38190928	0.568823300	2.429418	1.595991e-02
40	1.98879058	0.832620808	2.388591	1.779418e-02
41	1.24648398	0.537349850	2.319688	2.131399e-02
42	-1.02297342	0.448548889	-2.280629	2.356920e-02
43	0.53121677	0.235530302	2.255407	2.513387e-02
44	0.94552517	0.445314993	2.123273	3.489593e-02
45	0.75876947	0.359734950	2.109246	3.610237e-02
46	0.31505552	0.153902038	2.047117	4.188585e-02
47	0.60230297	0.303006100	1.987759	4.813135e-02
48	-1.72398341	0.633351107	-2.722003	7.053464e-03
49	-1.73162985	0.672552672	-2.574713	1.080288e-02
50	1.40517527	0.577484143	2.433271	1.584199e-02
51	1.06084803	0.484370055	2.190160	2.962496e-02
52	-1.19587595	0.551098554	-2.169986	3.112540e-02
53	-1.64071882	0.765922691	-2.142147	3.335640e-02
54	-1.00023779	0.488229883	-2.048703	4.172917e-02
55	2.63672296	1.094483296	2.409103	1.685057e-02
56	2.45015841	1.080854165	2.266872	2.441172e-02
57	-4.30619631	1.911648298	-2.252609	2.531293e-02
58	-3.03533294	1.354678028	-2.240631	2.609204e-02
59	-1.30220790	0.632918731	-2.057465	4.087106e-02
60	2.64126877	1.338047440	1.973972	4.968966e-02
61	-3.67116478	1.110737405	-3.305160	1.134557e-03
62	3.44767495	1.290913251	2.670726	8.233292e-03
63	2.77143392	1.133284577	2.445488	1.535345e-02
64	-2.16446255	0.913203166	-2.370187	1.869121e-02
65	-3.04553330	1.310471919	-2.323997	2.114431e-02
66	-1.41499552	0.623392502	-2.269831	2.422917e-02
67	2.84898428	1.322086066	2.154916	3.269775e-02
68	2.06378035	1.033782921	1.996338	4.740950e-02
69	-1.60388298	0.501460606	-3.198423	1.598027e-03
70	3.61480658	1.189030970	3.040128	2.866079e-03
71	3.41705215	1.183112602	2.888188	4.385132e-03
72	1.15001487	0.503589366	2.283636	2.374130e-02
73	-0.56940851	0.257053952	-2.215132	2.790237e-02
74	-0.54086285	0.259591953	-2.083512	3.849822e-02
75	2.81611022	1.081366124	2.604215	9.863034e-03
76	2.89446403	1.128748932	2.564312	1.103282e-02
77	-2.67399971	1.205215653	-2.218690	2.757337e-02
78	1.93526430	0.911786046	2.122498	3.496163e-02

79	-2.38204455	1.142778150	-2.084433	3.832432e-02
80	-0.98546785	0.473697019	-2.080376	3.869855e-02
81	-0.51027070	0.246217173	-2.072442	3.943945e-02
82	-3.25508719	1.612626661	-2.018500	4.480488e-02
83	1.83059622	0.923283297	1.982703	4.869798e-02
84	1.92662354	0.974892559	1.976242	4.943024e-02
85	-63.11228061	18.299143916	-3.448920	6.961397e-04
86	-18.99560267	6.474454880	-2.933931	3.724519e-03
87	-37.57126529	13.136276925	-2.860115	4.661362e-03
88	-20.87799165	7.877012247	-2.650496	8.750466e-03
89	-34.29253437	13.396960460	-2.559725	1.118154e-02
90	18.00487624	7.033716267	2.559796	1.125933e-02
91	-22.24640136	8.716342531	-2.552263	1.145816e-02
92	35.63252525	14.339975933	2.484839	1.389529e-02
93	-21.31391841	8.696128422	-2.450966	1.518276e-02
94	9.22836606	4.025048313	2.292734	2.288500e-02
95	-43.67920372	19.825923040	-2.203136	2.871405e-02
96	32.44233144	14.885182455	2.179505	3.058457e-02
97	8.19032939	3.984722059	2.055433	4.112841e-02
98	38.28092081	18.644969535	2.053150	4.174475e-02
99	32.47736732	16.058670424	2.022419	4.450233e-02
100	-3.03893495	0.976106550	-3.113323	2.106437e-03
101	-3.35377745	1.125067668	-2.980956	3.210955e-03
102	-1.71323841	0.615071881	-2.785428	5.832339e-03
103	-3.97885996	1.527245256	-2.605253	9.834164e-03
104	-2.76626207	1.242693663	-2.226021	2.707048e-02
105	1.50214332	0.760011616	1.976474	4.940376e-02
106	2.01612507	0.560745080	3.595440	4.048369e-04
107	1.81119209	0.505609409	3.582196	4.267781e-04
108	2.45760083	0.719848715	3.414052	7.675696e-04
109	-3.78364400	1.251889404	-3.022347	2.824222e-03
110	2.73484457	0.942579891	2.901446	4.109863e-03
111	-1.50396362	0.519354482	-2.895833	4.183270e-03
112	-3.11297409	1.138533957	-2.734195	6.789661e-03
113	3.31105546	1.216035462	2.722828	7.025352e-03
114	1.29999793	0.500259868	2.598645	1.001961e-02
115	-1.62032598	0.670295970	-2.417329	1.648719e-02
116	-4.26186641	1.789576549	-2.381494	1.813184e-02
117	1.51994588	0.671971767	2.261919	2.472535e-02
118	0.72616781	0.323555892	2.244335	2.585006e-02
119	1.69426463	0.763586920	2.218824	2.756466e-02
120	2.09008214	0.985079822	2.121739	3.502898e-02
121	1.94802594	0.932665512	2.088665	3.795004e-02

122	1.30612749	0.641814336	2.035055	4.310650e-02
123	-2.64232771	1.303962083	-2.026384	4.405776e-02
124	2.52370711	1.256586641	2.008383	4.588479e-02
125	-2.44065776	0.536392005	-4.550138	9.043370e-06
126	5.67543667	1.884232270	3.012068	2.911728e-03
127	-4.75004808	1.625973691	-2.921356	3.864292e-03
128	-2.23113065	0.866993508	-2.573411	1.075565e-02
129	2.41369365	0.956808582	2.522651	1.238513e-02
130	1.31655559	0.575021670	2.289576	2.303498e-02
131	1.46246094	0.662791077	2.206519	2.842620e-02
132	-3.40332414	1.601794500	-2.124695	3.477559e-02
133	-0.52850840	0.248981024	-2.122685	3.494574e-02
134	1.08845452	0.517128756	2.104804	3.649184e-02
135	1.44006702	0.693206136	2.077401	3.897495e-02
136	1.67281877	0.818385855	2.044047	4.219111e-02
137	-0.68543664	0.336643049	-2.036093	4.299055e-02
138	1.25364557	0.628433281	1.994875	4.734331e-02
139	-1.88204560	0.469592238	-4.007829	8.497548e-05
140	2.12169981	0.837652090	2.532913	1.203896e-02
141	-2.90900041	1.166014153	-2.494824	1.336888e-02
142	1.41434768	0.580250157	2.437479	1.561818e-02
143	1.21179202	0.503411144	2.407162	1.693794e-02
144	2.56440351	1.089284373	2.354209	1.948069e-02
145	1.02709306	0.452727945	2.268676	2.429977e-02
146	1.45098364	0.648113997	2.238778	2.621435e-02
147	1.41876373	0.716467886	1.980220	4.897832e-02
148	-2.77281742	1.402314459	-1.977315	4.930797e-02
149	-0.48157729	0.244083599	-1.973001	4.980101e-02

```
# WHO-5 -----
m_WHO5_cross <- lmer(
  WHO5_total ~ Age + Gender + Role + EmpStatus + OftenTeach + Formal + Qualified +
    (1 | EmpCount),
  data = score_df, REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_WHO5_cross)
```

Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's method [lmerModLmerTest]

Formula: WHO5_total ~ Age + Gender + Role + EmpStatus + OftenTeach + Formal + Qualified + (1 | EmpCount)

```
Data: score_df
Control: lmerControl(optimizer = "bobyqa")
```

AIC	BIC	logLik	-2*log(L)	df.resid
1874.7	2015.5	-895.3	1790.7	169

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.6973	-0.6172	0.0000	0.6095	2.4905

Random effects:

Groups	Name	Variance	Std.Dev.
EmpCount	(Intercept)	41.16	6.415
Residual		263.66	16.238

Number of obs: 211, groups: EmpCount, 29

Fixed effects:

(Intercept)

Age

GenderMale

RoleClassroom teacher / Other

RoleClassroom teacher / Principal/Coordinator

RoleClassroom teacher / Subject Specialist

RoleClassroom teacher / Subject Specialist / Other

RoleClassroom teacher / Subject Specialist / Principal/Coordinator

RoleClassroom teacher / Subject Specialist / Support/Special

RoleClassroom teacher / Subject Specialist / Teaching Assistant

RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other

RoleClassroom teacher / Support/Special

RoleClassroom teacher / Support/Special / Principal/Coordinator

RoleClassroom teacher / Teaching Assistant

RoleOther

RolePrincipal/Coordinator

RoleSubject Specialist

RoleSubject Specialist / Other

RoleSubject Specialist / Principal/Coordinator

RoleSubject Specialist / Support/Special

RoleSubject Specialist / Support/Special / Teaching Assistant

RoleSubject Specialist / Teaching Assistant / Other

RoleSupport/Special

RoleSupport/Special / Principal/Coordinator / Other

RoleSupport/Special / Teaching Assistant

RoleTeaching Assistant
 EmpStatusFreelance/Self employed
 EmpStatusOther
 EmpStatusPermanent, Full
 EmpStatusPermanent, Part
 OftenTeachIrregular/variable schedule
 OftenTeachLess than once per week
 OftenTeachOnce per week
 OftenTeachSeveral times per week
 FormalNo
 FormalYes
 QualifiedDoctorate
 QualifiedMaster's degree
 QualifiedOther
 QualifiedProfessional Qualification

(Intercept)

Age
 GenderMale
 RoleClassroom teacher / Other
 RoleClassroom teacher / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist
 RoleClassroom teacher / Subject Specialist / Other
 RoleClassroom teacher / Subject Specialist / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist / Support/Special
 RoleClassroom teacher / Subject Specialist / Teaching Assistant
 RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
 RoleClassroom teacher / Support/Special
 RoleClassroom teacher / Support/Special / Principal/Coordinator
 RoleClassroom teacher / Teaching Assistant
 RoleOther
 RolePrincipal/Coordinator
 RoleSubject Specialist
 RoleSubject Specialist / Other
 RoleSubject Specialist / Principal/Coordinator
 RoleSubject Specialist / Support/Special
 RoleSubject Specialist / Support/Special / Teaching Assistant
 RoleSubject Specialist / Teaching Assistant / Other
 RoleSupport/Special
 RoleSupport/Special / Principal/Coordinator / Other
 RoleSupport/Special / Teaching Assistant
 RoleTeaching Assistant
 EmpStatusFreelance/Self employed

EmpStatusOther
 EmpStatusPermanent, Full
 EmpStatusPermanent, Part
 OftenTeachIrregular/variable schedule
 OftenTeachLess than once per week
 OftenTeachOnce per week
 OftenTeachSeveral times per week
 FormalNo
 FormalYes
 QualifiedDoctorate
 QualifiedMaster's degree
 QualifiedOther
 QualifiedProfessional Qualification

 (Intercept)
 Age
 GenderMale
 RoleClassroom teacher / Other
 RoleClassroom teacher / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist
 RoleClassroom teacher / Subject Specialist / Other
 RoleClassroom teacher / Subject Specialist / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist / Support/Special
 RoleClassroom teacher / Subject Specialist / Teaching Assistant
 RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
 RoleClassroom teacher / Support/Special
 RoleClassroom teacher / Support/Special / Principal/Coordinator
 RoleClassroom teacher / Teaching Assistant
 RoleOther
 RolePrincipal/Coordinator
 RoleSubject Specialist
 RoleSubject Specialist / Other
 RoleSubject Specialist / Principal/Coordinator
 RoleSubject Specialist / Support/Special
 RoleSubject Specialist / Support/Special / Teaching Assistant
 RoleSubject Specialist / Teaching Assistant / Other
 RoleSupport/Special
 RoleSupport/Special / Principal/Coordinator / Other
 RoleSupport/Special / Teaching Assistant
 RoleTeaching Assistant
 EmpStatusFreelance/Self employed
 EmpStatusOther
 EmpStatusPermanent, Full

EmpStatusPermanent, Part
 OftenTeachIrregular/variable schedule
 OftenTeachLess than once per week
 OftenTeachOnce per week
 OftenTeachSeveral times per week
 FormalNo
 FormalYes
 QualifiedDoctorate
 QualifiedMaster's degree
 QualifiedOther
 QualifiedProfessional Qualification

 (Intercept)
 Age
 GenderMale
 RoleClassroom teacher / Other
 RoleClassroom teacher / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist
 RoleClassroom teacher / Subject Specialist / Other
 RoleClassroom teacher / Subject Specialist / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist / Support/Special
 RoleClassroom teacher / Subject Specialist / Teaching Assistant
 RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
 RoleClassroom teacher / Support/Special
 RoleClassroom teacher / Support/Special / Principal/Coordinator
 RoleClassroom teacher / Teaching Assistant
 RoleOther
 RolePrincipal/Coordinator
 RoleSubject Specialist
 RoleSubject Specialist / Other
 RoleSubject Specialist / Principal/Coordinator
 RoleSubject Specialist / Support/Special
 RoleSubject Specialist / Support/Special / Teaching Assistant
 RoleSubject Specialist / Teaching Assistant / Other
 RoleSupport/Special
 RoleSupport/Special / Principal/Coordinator / Other
 RoleSupport/Special / Teaching Assistant
 RoleTeaching Assistant
 EmpStatusFreelance/Self employed
 EmpStatusOther
 EmpStatusPermanent, Full
 EmpStatusPermanent, Part
 OftenTeachIrregular/variable schedule

OftenTeachLess than once per week
OftenTeachOnce per week
OftenTeachSeveral times per week
FormalNo
FormalYes
QualifiedDoctorate
QualifiedMaster's degree
QualifiedOther
QualifiedProfessional Qualification

(Intercept)

Age

GenderMale

RoleClassroom teacher / Other

RoleClassroom teacher / Principal/Coordinator

RoleClassroom teacher / Subject Specialist

RoleClassroom teacher / Subject Specialist / Other

RoleClassroom teacher / Subject Specialist / Principal/Coordinator

RoleClassroom teacher / Subject Specialist / Support/Special

RoleClassroom teacher / Subject Specialist / Teaching Assistant

RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other

RoleClassroom teacher / Support/Special

RoleClassroom teacher / Support/Special / Principal/Coordinator

RoleClassroom teacher / Teaching Assistant

RoleOther

RolePrincipal/Coordinator

RoleSubject Specialist

RoleSubject Specialist / Other

RoleSubject Specialist / Principal/Coordinator

RoleSubject Specialist / Support/Special

RoleSubject Specialist / Support/Special / Teaching Assistant

RoleSubject Specialist / Teaching Assistant / Other

RoleSupport/Special

RoleSupport/Special / Principal/Coordinator / Other

RoleSupport/Special / Teaching Assistant

RoleTeaching Assistant

EmpStatusFreelance/Self employed

EmpStatusOther

EmpStatusPermanent, Full

EmpStatusPermanent, Part

OftenTeachIrregular/variable schedule

OftenTeachLess than once per week

OftenTeachOnce per week

OftenTeachSeveral times per week
FormalNo
FormalYes
QualifiedDoctorate
QualifiedMaster's degree
QualifiedOther
QualifiedProfessional Qualification

(Intercept)

Age
GenderMale
RoleClassroom teacher / Other
RoleClassroom teacher / Principal/Coordinator
RoleClassroom teacher / Subject Specialist
RoleClassroom teacher / Subject Specialist / Other
RoleClassroom teacher / Subject Specialist / Principal/Coordinator
RoleClassroom teacher / Subject Specialist / Support/Special
RoleClassroom teacher / Subject Specialist / Teaching Assistant
RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
RoleClassroom teacher / Support/Special
RoleClassroom teacher / Support/Special / Principal/Coordinator
RoleClassroom teacher / Teaching Assistant
RoleOther
RolePrincipal/Coordinator
RoleSubject Specialist
RoleSubject Specialist / Other
RoleSubject Specialist / Principal/Coordinator
RoleSubject Specialist / Support/Special
RoleSubject Specialist / Support/Special / Teaching Assistant
RoleSubject Specialist / Teaching Assistant / Other
RoleSupport/Special
RoleSupport/Special / Principal/Coordinator / Other
RoleSupport/Special / Teaching Assistant
RoleTeaching Assistant
EmpStatusFreelance/Self employed
EmpStatusOther
EmpStatusPermanent, Full
EmpStatusPermanent, Part
OftenTeachIrregular/variable schedule
OftenTeachLess than once per week
OftenTeachOnce per week
OftenTeachSeveral times per week
FormalNo

```

FormalYes
QualifiedDoctorate
QualifiedMaster's degree
QualifiedOther
QualifiedProfessional Qualification
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Correlation matrix not shown by default, as $p = 40 > 12$.

```

Use print(x, correlation=TRUE) or
      vcov(x)          if you need it

```

```

# LoiTeS full + factors -----
m_LoiTeS_full <- lmer(
  Michele_full ~ Age + Gender + Role + EmpStatus + OftenTeach + Formal + Qualified +
    (1 | EmpCount),
  data = score_df, REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_LoiTeS_full)

```

Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's method [lmerModLmerTest]

```

Formula: Michele_full ~ Age + Gender + Role + EmpStatus + OftenTeach +
  Formal + Qualified + (1 | EmpCount)

```

```
Data: score_df
```

```
Control: lmerControl(optimizer = "bobyqa")
```

AIC	BIC	logLik	-2*log(L)	df.resid
436.5	577.3	-176.2	352.5	169

Scaled residuals:

Min	1Q	Median	3Q	Max
-3.14832	-0.56439	0.01094	0.60857	2.29543

Random effects:

Groups	Name	Variance	Std.Dev.
EmpCount	(Intercept)	0.0198	0.1407
	Residual	0.2987	0.5466

Number of obs: 211, groups: EmpCount, 29

Fixed effects:

(Intercept)

Age

GenderMale

RoleClassroom teacher / Other

RoleClassroom teacher / Principal/Coordinator

RoleClassroom teacher / Subject Specialist

RoleClassroom teacher / Subject Specialist / Other

RoleClassroom teacher / Subject Specialist / Principal/Coordinator

RoleClassroom teacher / Subject Specialist / Support/Special

RoleClassroom teacher / Subject Specialist / Teaching Assistant

RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other

RoleClassroom teacher / Support/Special

RoleClassroom teacher / Support/Special / Principal/Coordinator

RoleClassroom teacher / Teaching Assistant

RoleOther

RolePrincipal/Coordinator

RoleSubject Specialist

RoleSubject Specialist / Other

RoleSubject Specialist / Principal/Coordinator

RoleSubject Specialist / Support/Special

RoleSubject Specialist / Support/Special / Teaching Assistant

RoleSubject Specialist / Teaching Assistant / Other

RoleSupport/Special

RoleSupport/Special / Principal/Coordinator / Other

RoleSupport/Special / Teaching Assistant

RoleTeaching Assistant

EmpStatusFreelance/Self employed

EmpStatusOther

EmpStatusPermanent, Full

EmpStatusPermanent, Part

OftenTeachIrregular/variable schedule

OftenTeachLess than once per week

OftenTeachOnce per week

OftenTeachSeveral times per week

FormalNo

FormalYes

QualifiedDoctorate

QualifiedMaster's degree

QualifiedOther

QualifiedProfessional Qualification

(Intercept)

Age

GenderMale

RoleClassroom teacher / Other

RoleClassroom teacher / Principal/Coordinator

RoleClassroom teacher / Subject Specialist

RoleClassroom teacher / Subject Specialist / Other

RoleClassroom teacher / Subject Specialist / Principal/Coordinator

RoleClassroom teacher / Subject Specialist / Support/Special

RoleClassroom teacher / Subject Specialist / Teaching Assistant

RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other

RoleClassroom teacher / Support/Special

RoleClassroom teacher / Support/Special / Principal/Coordinator

RoleClassroom teacher / Teaching Assistant

RoleOther

RolePrincipal/Coordinator

RoleSubject Specialist

RoleSubject Specialist / Other

RoleSubject Specialist / Principal/Coordinator

RoleSubject Specialist / Support/Special

RoleSubject Specialist / Support/Special / Teaching Assistant

RoleSubject Specialist / Teaching Assistant / Other

RoleSupport/Special

RoleSupport/Special / Principal/Coordinator / Other

RoleSupport/Special / Teaching Assistant

RoleTeaching Assistant

EmpStatusFreelance/Self employed

EmpStatusOther

EmpStatusPermanent, Full

EmpStatusPermanent, Part

OftenTeachIrregular/variable schedule

OftenTeachLess than once per week

OftenTeachOnce per week

OftenTeachSeveral times per week

FormalNo

FormalYes

QualifiedDoctorate

QualifiedMaster's degree

QualifiedOther

QualifiedProfessional Qualification

(Intercept)

Age

GenderMale
 RoleClassroom teacher / Other
 RoleClassroom teacher / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist
 RoleClassroom teacher / Subject Specialist / Other
 RoleClassroom teacher / Subject Specialist / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist / Support/Special
 RoleClassroom teacher / Subject Specialist / Teaching Assistant
 RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
 RoleClassroom teacher / Support/Special
 RoleClassroom teacher / Support/Special / Principal/Coordinator
 RoleClassroom teacher / Teaching Assistant
 RoleOther
 RolePrincipal/Coordinator
 RoleSubject Specialist
 RoleSubject Specialist / Other
 RoleSubject Specialist / Principal/Coordinator
 RoleSubject Specialist / Support/Special
 RoleSubject Specialist / Support/Special / Teaching Assistant
 RoleSubject Specialist / Teaching Assistant / Other
 RoleSupport/Special
 RoleSupport/Special / Principal/Coordinator / Other
 RoleSupport/Special / Teaching Assistant
 RoleTeaching Assistant
 EmpStatusFreelance/Self employed
 EmpStatusOther
 EmpStatusPermanent, Full
 EmpStatusPermanent, Part
 OftenTeachIrregular/variable schedule
 OftenTeachLess than once per week
 OftenTeachOnce per week
 OftenTeachSeveral times per week
 FormalNo
 FormalYes
 QualifiedDoctorate
 QualifiedMaster's degree
 QualifiedOther
 QualifiedProfessional Qualification

 (Intercept)
 Age
 GenderMale
 RoleClassroom teacher / Other

RoleClassroom teacher / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist
 RoleClassroom teacher / Subject Specialist / Other
 RoleClassroom teacher / Subject Specialist / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist / Support/Special
 RoleClassroom teacher / Subject Specialist / Teaching Assistant
 RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
 RoleClassroom teacher / Support/Special
 RoleClassroom teacher / Support/Special / Principal/Coordinator
 RoleClassroom teacher / Teaching Assistant
 RoleOther
 RolePrincipal/Coordinator
 RoleSubject Specialist
 RoleSubject Specialist / Other
 RoleSubject Specialist / Principal/Coordinator
 RoleSubject Specialist / Support/Special
 RoleSubject Specialist / Support/Special / Teaching Assistant
 RoleSubject Specialist / Teaching Assistant / Other
 RoleSupport/Special
 RoleSupport/Special / Principal/Coordinator / Other
 RoleSupport/Special / Teaching Assistant
 RoleTeaching Assistant
 EmpStatusFreelance/Self employed
 EmpStatusOther
 EmpStatusPermanent, Full
 EmpStatusPermanent, Part
 OftenTeachIrregular/variable schedule
 OftenTeachLess than once per week
 OftenTeachOnce per week
 OftenTeachSeveral times per week
 FormalNo
 FormalYes
 QualifiedDoctorate
 QualifiedMaster's degree
 QualifiedOther
 QualifiedProfessional Qualification

 (Intercept)
 Age
 GenderMale
 RoleClassroom teacher / Other
 RoleClassroom teacher / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist

RoleClassroom teacher / Subject Specialist / Other
 RoleClassroom teacher / Subject Specialist / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist / Support/Special
 RoleClassroom teacher / Subject Specialist / Teaching Assistant
 RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
 RoleClassroom teacher / Support/Special
 RoleClassroom teacher / Support/Special / Principal/Coordinator
 RoleClassroom teacher / Teaching Assistant
 RoleOther
 RolePrincipal/Coordinator
 RoleSubject Specialist
 RoleSubject Specialist / Other
 RoleSubject Specialist / Principal/Coordinator
 RoleSubject Specialist / Support/Special
 RoleSubject Specialist / Support/Special / Teaching Assistant
 RoleSubject Specialist / Teaching Assistant / Other
 RoleSupport/Special
 RoleSupport/Special / Principal/Coordinator / Other
 RoleSupport/Special / Teaching Assistant
 RoleTeaching Assistant
 EmpStatusFreelance/Self employed
 EmpStatusOther
 EmpStatusPermanent, Full
 EmpStatusPermanent, Part
 OftenTeachIrregular/variable schedule
 OftenTeachLess than once per week
 OftenTeachOnce per week
 OftenTeachSeveral times per week
 FormalNo
 FormalYes
 QualifiedDoctorate
 QualifiedMaster's degree
 QualifiedOther
 QualifiedProfessional Qualification

 (Intercept)
 Age
 GenderMale
 RoleClassroom teacher / Other
 RoleClassroom teacher / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist
 RoleClassroom teacher / Subject Specialist / Other
 RoleClassroom teacher / Subject Specialist / Principal/Coordinator

```

RoleClassroom teacher / Subject Specialist / Support/Special
RoleClassroom teacher / Subject Specialist / Teaching Assistant
RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
RoleClassroom teacher / Support/Special
RoleClassroom teacher / Support/Special / Principal/Coordinator
RoleClassroom teacher / Teaching Assistant
RoleOther
RolePrincipal/Coordinator
RoleSubject Specialist
RoleSubject Specialist / Other
RoleSubject Specialist / Principal/Coordinator
RoleSubject Specialist / Support/Special
RoleSubject Specialist / Support/Special / Teaching Assistant
RoleSubject Specialist / Teaching Assistant / Other
RoleSupport/Special
RoleSupport/Special / Principal/Coordinator / Other
RoleSupport/Special / Teaching Assistant
RoleTeaching Assistant
EmpStatusFreelance/Self employed
EmpStatusOther
EmpStatusPermanent, Full
EmpStatusPermanent, Part
OftenTeachIrregular/variable schedule
OftenTeachLess than once per week
OftenTeachOnce per week
OftenTeachSeveral times per week
FormalNo
FormalYes
QualifiedDoctorate
QualifiedMaster's degree
QualifiedOther
QualifiedProfessional Qualification
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Correlation matrix not shown by default, as $p = 40 > 12$.
 Use `print(x, correlation=TRUE)` or
`vcov(x)` if you need it

```

m_LoiTeS_F1 <- lmer(
  Michele_F1 ~ Age + Gender + Role + EmpStatus + OftenTeach + Formal + Qualified +

```

```

    (1 | EmpCount),
    data = score_df, REML = FALSE,
    control = lmerControl(optimizer = "bobyqa")
)

```

boundary (singular) fit: see help('isSingular')

```

m_LoiTeS_F2 <- lmer(
  Michele_F2 ~ Age + Gender + Role + EmpStatus + OftenTeach + Formal + Qualified +
    (1 | EmpCount),
  data = score_df, REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)

m_LoiTeS_F3 <- lmer(
  Michele_F3 ~ Age + Gender + Role + EmpStatus + OftenTeach + Formal + Qualified +
    (1 | EmpCount),
  data = score_df, REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)

```

boundary (singular) fit: see help('isSingular')

```

# Hughes (R-UCLA) -----
m_Hughes <- lmer(
  Hughes_total ~ Age + Gender + Role + EmpStatus + OftenTeach + Formal + Qualified +
    (1 | EmpCount),
  data = score_df, REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)

```

boundary (singular) fit: see help('isSingular')

```
summary(m_Hughes)
```

```

Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
method [lmerModLmerTest]
Formula: Hughes_total ~ Age + Gender + Role + EmpStatus + OftenTeach +
  Formal + Qualified + (1 | EmpCount)

```

```
Data: score_df
Control: lmerControl(optimizer = "bobyqa")
```

AIC	BIC	logLik	-2*log(L)	df.resid
1649.4	1790.2	-782.7	1565.4	169

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.13995	-0.60593	-0.06947	0.68494	2.37737

Random effects:

Groups	Name	Variance	Std.Dev.
EmpCount	(Intercept)	1.362e-13	3.691e-07
Residual		9.764e+01	9.881e+00

Number of obs: 211, groups: EmpCount, 29

Fixed effects:

(Intercept)

Age

GenderMale

RoleClassroom teacher / Other

RoleClassroom teacher / Principal/Coordinator

RoleClassroom teacher / Subject Specialist

RoleClassroom teacher / Subject Specialist / Other

RoleClassroom teacher / Subject Specialist / Principal/Coordinator

RoleClassroom teacher / Subject Specialist / Support/Special

RoleClassroom teacher / Subject Specialist / Teaching Assistant

RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other

RoleClassroom teacher / Support/Special

RoleClassroom teacher / Support/Special / Principal/Coordinator

RoleClassroom teacher / Teaching Assistant

RoleOther

RolePrincipal/Coordinator

RoleSubject Specialist

RoleSubject Specialist / Other

RoleSubject Specialist / Principal/Coordinator

RoleSubject Specialist / Support/Special

RoleSubject Specialist / Support/Special / Teaching Assistant

RoleSubject Specialist / Teaching Assistant / Other

RoleSupport/Special

RoleSupport/Special / Principal/Coordinator / Other

RoleSupport/Special / Teaching Assistant

RoleTeaching Assistant
 EmpStatusFreelance/Self employed
 EmpStatusOther
 EmpStatusPermanent, Full
 EmpStatusPermanent, Part
 OftenTeachIrregular/variable schedule
 OftenTeachLess than once per week
 OftenTeachOnce per week
 OftenTeachSeveral times per week
 FormalNo
 FormalYes
 QualifiedDoctorate
 QualifiedMaster's degree
 QualifiedOther
 QualifiedProfessional Qualification

(Intercept)

Age
 GenderMale
 RoleClassroom teacher / Other
 RoleClassroom teacher / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist
 RoleClassroom teacher / Subject Specialist / Other
 RoleClassroom teacher / Subject Specialist / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist / Support/Special
 RoleClassroom teacher / Subject Specialist / Teaching Assistant
 RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
 RoleClassroom teacher / Support/Special
 RoleClassroom teacher / Support/Special / Principal/Coordinator
 RoleClassroom teacher / Teaching Assistant
 RoleOther
 RolePrincipal/Coordinator
 RoleSubject Specialist
 RoleSubject Specialist / Other
 RoleSubject Specialist / Principal/Coordinator
 RoleSubject Specialist / Support/Special
 RoleSubject Specialist / Support/Special / Teaching Assistant
 RoleSubject Specialist / Teaching Assistant / Other
 RoleSupport/Special
 RoleSupport/Special / Principal/Coordinator / Other
 RoleSupport/Special / Teaching Assistant
 RoleTeaching Assistant
 EmpStatusFreelance/Self employed

EmpStatusOther
 EmpStatusPermanent, Full
 EmpStatusPermanent, Part
 OftenTeachIrregular/variable schedule
 OftenTeachLess than once per week
 OftenTeachOnce per week
 OftenTeachSeveral times per week
 FormalNo
 FormalYes
 QualifiedDoctorate
 QualifiedMaster's degree
 QualifiedOther
 QualifiedProfessional Qualification

 (Intercept)
 Age
 GenderMale
 RoleClassroom teacher / Other
 RoleClassroom teacher / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist
 RoleClassroom teacher / Subject Specialist / Other
 RoleClassroom teacher / Subject Specialist / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist / Support/Special
 RoleClassroom teacher / Subject Specialist / Teaching Assistant
 RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
 RoleClassroom teacher / Support/Special
 RoleClassroom teacher / Support/Special / Principal/Coordinator
 RoleClassroom teacher / Teaching Assistant
 RoleOther
 RolePrincipal/Coordinator
 RoleSubject Specialist
 RoleSubject Specialist / Other
 RoleSubject Specialist / Principal/Coordinator
 RoleSubject Specialist / Support/Special
 RoleSubject Specialist / Support/Special / Teaching Assistant
 RoleSubject Specialist / Teaching Assistant / Other
 RoleSupport/Special
 RoleSupport/Special / Principal/Coordinator / Other
 RoleSupport/Special / Teaching Assistant
 RoleTeaching Assistant
 EmpStatusFreelance/Self employed
 EmpStatusOther
 EmpStatusPermanent, Full

EmpStatusPermanent, Part
 OftenTeachIrregular/variable schedule
 OftenTeachLess than once per week
 OftenTeachOnce per week
 OftenTeachSeveral times per week
 FormalNo
 FormalYes
 QualifiedDoctorate
 QualifiedMaster's degree
 QualifiedOther
 QualifiedProfessional Qualification

 (Intercept)
 Age
 GenderMale
 RoleClassroom teacher / Other
 RoleClassroom teacher / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist
 RoleClassroom teacher / Subject Specialist / Other
 RoleClassroom teacher / Subject Specialist / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist / Support/Special
 RoleClassroom teacher / Subject Specialist / Teaching Assistant
 RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
 RoleClassroom teacher / Support/Special
 RoleClassroom teacher / Support/Special / Principal/Coordinator
 RoleClassroom teacher / Teaching Assistant
 RoleOther
 RolePrincipal/Coordinator
 RoleSubject Specialist
 RoleSubject Specialist / Other
 RoleSubject Specialist / Principal/Coordinator
 RoleSubject Specialist / Support/Special
 RoleSubject Specialist / Support/Special / Teaching Assistant
 RoleSubject Specialist / Teaching Assistant / Other
 RoleSupport/Special
 RoleSupport/Special / Principal/Coordinator / Other
 RoleSupport/Special / Teaching Assistant
 RoleTeaching Assistant
 EmpStatusFreelance/Self employed
 EmpStatusOther
 EmpStatusPermanent, Full
 EmpStatusPermanent, Part
 OftenTeachIrregular/variable schedule

OftenTeachLess than once per week
OftenTeachOnce per week
OftenTeachSeveral times per week
FormalNo
FormalYes
QualifiedDoctorate
QualifiedMaster's degree
QualifiedOther
QualifiedProfessional Qualification

(Intercept)

Age
GenderMale
RoleClassroom teacher / Other
RoleClassroom teacher / Principal/Coordinator
RoleClassroom teacher / Subject Specialist
RoleClassroom teacher / Subject Specialist / Other
RoleClassroom teacher / Subject Specialist / Principal/Coordinator
RoleClassroom teacher / Subject Specialist / Support/Special
RoleClassroom teacher / Subject Specialist / Teaching Assistant
RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
RoleClassroom teacher / Support/Special
RoleClassroom teacher / Support/Special / Principal/Coordinator
RoleClassroom teacher / Teaching Assistant
RoleOther
RolePrincipal/Coordinator
RoleSubject Specialist
RoleSubject Specialist / Other
RoleSubject Specialist / Principal/Coordinator
RoleSubject Specialist / Support/Special
RoleSubject Specialist / Support/Special / Teaching Assistant
RoleSubject Specialist / Teaching Assistant / Other
RoleSupport/Special
RoleSupport/Special / Principal/Coordinator / Other
RoleSupport/Special / Teaching Assistant
RoleTeaching Assistant
EmpStatusFreelance/Self employed
EmpStatusOther
EmpStatusPermanent, Full
EmpStatusPermanent, Part
OftenTeachIrregular/variable schedule
OftenTeachLess than once per week
OftenTeachOnce per week

OftenTeachSeveral times per week
FormalNo
FormalYes
QualifiedDoctorate
QualifiedMaster's degree
QualifiedOther
QualifiedProfessional Qualification

(Intercept)

Age
GenderMale
RoleClassroom teacher / Other
RoleClassroom teacher / Principal/Coordinator
RoleClassroom teacher / Subject Specialist
RoleClassroom teacher / Subject Specialist / Other
RoleClassroom teacher / Subject Specialist / Principal/Coordinator
RoleClassroom teacher / Subject Specialist / Support/Special
RoleClassroom teacher / Subject Specialist / Teaching Assistant
RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
RoleClassroom teacher / Support/Special
RoleClassroom teacher / Support/Special / Principal/Coordinator
RoleClassroom teacher / Teaching Assistant
RoleOther
RolePrincipal/Coordinator
RoleSubject Specialist
RoleSubject Specialist / Other
RoleSubject Specialist / Principal/Coordinator
RoleSubject Specialist / Support/Special
RoleSubject Specialist / Support/Special / Teaching Assistant
RoleSubject Specialist / Teaching Assistant / Other
RoleSupport/Special
RoleSupport/Special / Principal/Coordinator / Other
RoleSupport/Special / Teaching Assistant
RoleTeaching Assistant
EmpStatusFreelance/Self employed
EmpStatusOther
EmpStatusPermanent, Full
EmpStatusPermanent, Part
OftenTeachIrregular/variable schedule
OftenTeachLess than once per week
OftenTeachOnce per week
OftenTeachSeveral times per week
FormalNo

```

FormalYes
QualifiedDoctorate
QualifiedMaster's degree
QualifiedOther
QualifiedProfessional Qualification
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

Correlation matrix not shown by default, as p = 40 > 12.
Use print(x, correlation=TRUE) or
      vcov(x)          if you need it

```

```

optimizer (bobyqa) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

```

```

# Tschannen-Moran -----
m_Tschannen_total <- lmer(
  Tschannen_total ~ Age + Gender + Role + EmpStatus + OftenTeach + Formal + Qualified +
    (1 | EmpCount),
  data = score_df, REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)

```

```

boundary (singular) fit: see help('isSingular')

```

```
summary(m_Tschannen_total)
```

```

Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
method [lmerModLmerTest]
Formula: Tschannen_total ~ Age + Gender + Role + EmpStatus + OftenTeach +
  Formal + Qualified + (1 | EmpCount)
Data: score_df
Control: lmerControl(optimizer = "bobyqa")

```

AIC	BIC	logLik	-2*log(L)	df.resid
693.8	834.5	-304.9	609.8	169

```

Scaled residuals:
  Min       1Q   Median       3Q      Max

```

-3.4456 -0.6067 0.0000 0.6447 2.3324

Random effects:

Groups	Name	Variance	Std.Dev.
EmpCount	(Intercept)	0.000	0.000
Residual		1.053	1.026

Number of obs: 211, groups: EmpCount, 29

Fixed effects:

(Intercept)

Age

GenderMale

RoleClassroom teacher / Other

RoleClassroom teacher / Principal/Coordinator

RoleClassroom teacher / Subject Specialist

RoleClassroom teacher / Subject Specialist / Other

RoleClassroom teacher / Subject Specialist / Principal/Coordinator

RoleClassroom teacher / Subject Specialist / Support/Special

RoleClassroom teacher / Subject Specialist / Teaching Assistant

RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other

RoleClassroom teacher / Support/Special

RoleClassroom teacher / Support/Special / Principal/Coordinator

RoleClassroom teacher / Teaching Assistant

RoleOther

RolePrincipal/Coordinator

RoleSubject Specialist

RoleSubject Specialist / Other

RoleSubject Specialist / Principal/Coordinator

RoleSubject Specialist / Support/Special

RoleSubject Specialist / Support/Special / Teaching Assistant

RoleSubject Specialist / Teaching Assistant / Other

RoleSupport/Special

RoleSupport/Special / Principal/Coordinator / Other

RoleSupport/Special / Teaching Assistant

RoleTeaching Assistant

EmpStatusFreelance/Self employed

EmpStatusOther

EmpStatusPermanent, Full

EmpStatusPermanent, Part

OftenTeachIrregular/variable schedule

OftenTeachLess than once per week

OftenTeachOnce per week

OftenTeachSeveral times per week
FormalNo
FormalYes
QualifiedDoctorate
QualifiedMaster's degree
QualifiedOther
QualifiedProfessional Qualification

(Intercept)

Age
GenderMale
RoleClassroom teacher / Other
RoleClassroom teacher / Principal/Coordinator
RoleClassroom teacher / Subject Specialist
RoleClassroom teacher / Subject Specialist / Other
RoleClassroom teacher / Subject Specialist / Principal/Coordinator
RoleClassroom teacher / Subject Specialist / Support/Special
RoleClassroom teacher / Subject Specialist / Teaching Assistant
RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
RoleClassroom teacher / Support/Special
RoleClassroom teacher / Support/Special / Principal/Coordinator
RoleClassroom teacher / Teaching Assistant
RoleOther
RolePrincipal/Coordinator
RoleSubject Specialist
RoleSubject Specialist / Other
RoleSubject Specialist / Principal/Coordinator
RoleSubject Specialist / Support/Special
RoleSubject Specialist / Support/Special / Teaching Assistant
RoleSubject Specialist / Teaching Assistant / Other
RoleSupport/Special
RoleSupport/Special / Principal/Coordinator / Other
RoleSupport/Special / Teaching Assistant
RoleTeaching Assistant
EmpStatusFreelance/Self employed
EmpStatusOther
EmpStatusPermanent, Full
EmpStatusPermanent, Part
OftenTeachIrregular/variable schedule
OftenTeachLess than once per week
OftenTeachOnce per week
OftenTeachSeveral times per week
FormalNo

FormalYes
 QualifiedDoctorate
 QualifiedMaster's degree
 QualifiedOther
 QualifiedProfessional Qualification

 (Intercept)
 Age
 GenderMale
 RoleClassroom teacher / Other
 RoleClassroom teacher / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist
 RoleClassroom teacher / Subject Specialist / Other
 RoleClassroom teacher / Subject Specialist / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist / Support/Special
 RoleClassroom teacher / Subject Specialist / Teaching Assistant
 RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
 RoleClassroom teacher / Support/Special
 RoleClassroom teacher / Support/Special / Principal/Coordinator
 RoleClassroom teacher / Teaching Assistant
 RoleOther
 RolePrincipal/Coordinator
 RoleSubject Specialist
 RoleSubject Specialist / Other
 RoleSubject Specialist / Principal/Coordinator
 RoleSubject Specialist / Support/Special
 RoleSubject Specialist / Support/Special / Teaching Assistant
 RoleSubject Specialist / Teaching Assistant / Other
 RoleSupport/Special
 RoleSupport/Special / Principal/Coordinator / Other
 RoleSupport/Special / Teaching Assistant
 RoleTeaching Assistant
 EmpStatusFreelance/Self employed
 EmpStatusOther
 EmpStatusPermanent, Full
 EmpStatusPermanent, Part
 OftenTeachIrregular/variable schedule
 OftenTeachLess than once per week
 OftenTeachOnce per week
 OftenTeachSeveral times per week
 FormalNo
 FormalYes
 QualifiedDoctorate

QualifiedMaster's degree
 QualifiedOther
 QualifiedProfessional Qualification

 (Intercept)
 Age
 GenderMale
 RoleClassroom teacher / Other
 RoleClassroom teacher / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist
 RoleClassroom teacher / Subject Specialist / Other
 RoleClassroom teacher / Subject Specialist / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist / Support/Special
 RoleClassroom teacher / Subject Specialist / Teaching Assistant
 RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
 RoleClassroom teacher / Support/Special
 RoleClassroom teacher / Support/Special / Principal/Coordinator
 RoleClassroom teacher / Teaching Assistant
 RoleOther
 RolePrincipal/Coordinator
 RoleSubject Specialist
 RoleSubject Specialist / Other
 RoleSubject Specialist / Principal/Coordinator
 RoleSubject Specialist / Support/Special
 RoleSubject Specialist / Support/Special / Teaching Assistant
 RoleSubject Specialist / Teaching Assistant / Other
 RoleSupport/Special
 RoleSupport/Special / Principal/Coordinator / Other
 RoleSupport/Special / Teaching Assistant
 RoleTeaching Assistant
 EmpStatusFreelance/Self employed
 EmpStatusOther
 EmpStatusPermanent, Full
 EmpStatusPermanent, Part
 OftenTeachIrregular/variable schedule
 OftenTeachLess than once per week
 OftenTeachOnce per week
 OftenTeachSeveral times per week
 FormalNo
 FormalYes
 QualifiedDoctorate
 QualifiedMaster's degree
 QualifiedOther

QualifiedProfessional Qualification

(Intercept)

Age

GenderMale

RoleClassroom teacher / Other

RoleClassroom teacher / Principal/Coordinator

RoleClassroom teacher / Subject Specialist

RoleClassroom teacher / Subject Specialist / Other

RoleClassroom teacher / Subject Specialist / Principal/Coordinator

RoleClassroom teacher / Subject Specialist / Support/Special

RoleClassroom teacher / Subject Specialist / Teaching Assistant

RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other

RoleClassroom teacher / Support/Special

RoleClassroom teacher / Support/Special / Principal/Coordinator

RoleClassroom teacher / Teaching Assistant

RoleOther

RolePrincipal/Coordinator

RoleSubject Specialist

RoleSubject Specialist / Other

RoleSubject Specialist / Principal/Coordinator

RoleSubject Specialist / Support/Special

RoleSubject Specialist / Support/Special / Teaching Assistant

RoleSubject Specialist / Teaching Assistant / Other

RoleSupport/Special

RoleSupport/Special / Principal/Coordinator / Other

RoleSupport/Special / Teaching Assistant

RoleTeaching Assistant

EmpStatusFreelance/Self employed

EmpStatusOther

EmpStatusPermanent, Full

EmpStatusPermanent, Part

OftenTeachIrregular/variable schedule

OftenTeachLess than once per week

OftenTeachOnce per week

OftenTeachSeveral times per week

FormalNo

FormalYes

QualifiedDoctorate

QualifiedMaster's degree

QualifiedOther

QualifiedProfessional Qualification

```

(Intercept)
Age
GenderMale
RoleClassroom teacher / Other
RoleClassroom teacher / Principal/Coordinator
RoleClassroom teacher / Subject Specialist
RoleClassroom teacher / Subject Specialist / Other
RoleClassroom teacher / Subject Specialist / Principal/Coordinator
RoleClassroom teacher / Subject Specialist / Support/Special
RoleClassroom teacher / Subject Specialist / Teaching Assistant
RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
RoleClassroom teacher / Support/Special
RoleClassroom teacher / Support/Special / Principal/Coordinator
RoleClassroom teacher / Teaching Assistant
RoleOther
RolePrincipal/Coordinator
RoleSubject Specialist
RoleSubject Specialist / Other
RoleSubject Specialist / Principal/Coordinator
RoleSubject Specialist / Support/Special
RoleSubject Specialist / Support/Special / Teaching Assistant
RoleSubject Specialist / Teaching Assistant / Other
RoleSupport/Special
RoleSupport/Special / Principal/Coordinator / Other
RoleSupport/Special / Teaching Assistant
RoleTeaching Assistant
EmpStatusFreelance/Self employed
EmpStatusOther
EmpStatusPermanent, Full
EmpStatusPermanent, Part
OftenTeachIrregular/variable schedule
OftenTeachLess than once per week
OftenTeachOnce per week
OftenTeachSeveral times per week
FormalNo
FormalYes
QualifiedDoctorate
QualifiedMaster's degree
QualifiedOther
QualifiedProfessional Qualification
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Correlation matrix not shown by default, as $p = 40 > 12$.

Use `print(x, correlation=TRUE)` or

`vcov(x)` if you need it

optimizer (bobyqa) convergence code: 0 (OK)

boundary (singular) fit: see `help('isSingular')`

```
m_Tschannen_instr <- update(m_Tschannen_total, Tschannen_instruction ~ .)
```

boundary (singular) fit: see `help('isSingular')`

```
m_Tschannen_mgmt <- update(m_Tschannen_total, Tschannen_class_manager ~ .)
```

boundary (singular) fit: see `help('isSingular')`

```
m_Tschannen_eng <- update(m_Tschannen_total, Tschannen_engagement ~ .)
```

```
# Zimet MSPSS -----
m_Zimet_total <- lmer(
  Zimet_total ~ Age + Gender + Role + EmpStatus + OftenTeach + Formal + Qualified +
    (1 | EmpCount),
  data = score_df, REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)
```

boundary (singular) fit: see `help('isSingular')`

```
summary(m_Zimet_total)
```

Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's method [lmerModLmerTest]

Formula: `Zimet_total ~ Age + Gender + Role + EmpStatus + OftenTeach + Formal + Qualified + (1 | EmpCount)`

Data: `score_df`

Control: `lmerControl(optimizer = "bobyqa")`

AIC	BIC	logLik	-2*log(L)	df.resid
-----	-----	--------	-----------	----------

690.9 831.7 -303.5 606.9 169

Scaled residuals:

Min	1Q	Median	3Q	Max
-3.9434	-0.5655	0.1122	0.7289	1.6548

Random effects:

Groups	Name	Variance	Std.Dev.
EmpCount	(Intercept)	0.000	0.000
	Residual	1.039	1.019

Number of obs: 211, groups: EmpCount, 29

Fixed effects:

(Intercept)

Age

GenderMale

RoleClassroom teacher / Other

RoleClassroom teacher / Principal/Coordinator

RoleClassroom teacher / Subject Specialist

RoleClassroom teacher / Subject Specialist / Other

RoleClassroom teacher / Subject Specialist / Principal/Coordinator

RoleClassroom teacher / Subject Specialist / Support/Special

RoleClassroom teacher / Subject Specialist / Teaching Assistant

RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other

RoleClassroom teacher / Support/Special

RoleClassroom teacher / Support/Special / Principal/Coordinator

RoleClassroom teacher / Teaching Assistant

RoleOther

RolePrincipal/Coordinator

RoleSubject Specialist

RoleSubject Specialist / Other

RoleSubject Specialist / Principal/Coordinator

RoleSubject Specialist / Support/Special

RoleSubject Specialist / Support/Special / Teaching Assistant

RoleSubject Specialist / Teaching Assistant / Other

RoleSupport/Special

RoleSupport/Special / Principal/Coordinator / Other

RoleSupport/Special / Teaching Assistant

RoleTeaching Assistant

EmpStatusFreelance/Self employed

EmpStatusOther

EmpStatusPermanent, Full

EmpStatusPermanent, Part
 OftenTeachIrregular/variable schedule
 OftenTeachLess than once per week
 OftenTeachOnce per week
 OftenTeachSeveral times per week
 FormalNo
 FormalYes
 QualifiedDoctorate
 QualifiedMaster's degree
 QualifiedOther
 QualifiedProfessional Qualification

 (Intercept)
 Age
 GenderMale
 RoleClassroom teacher / Other
 RoleClassroom teacher / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist
 RoleClassroom teacher / Subject Specialist / Other
 RoleClassroom teacher / Subject Specialist / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist / Support/Special
 RoleClassroom teacher / Subject Specialist / Teaching Assistant
 RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
 RoleClassroom teacher / Support/Special
 RoleClassroom teacher / Support/Special / Principal/Coordinator
 RoleClassroom teacher / Teaching Assistant
 RoleOther
 RolePrincipal/Coordinator
 RoleSubject Specialist
 RoleSubject Specialist / Other
 RoleSubject Specialist / Principal/Coordinator
 RoleSubject Specialist / Support/Special
 RoleSubject Specialist / Support/Special / Teaching Assistant
 RoleSubject Specialist / Teaching Assistant / Other
 RoleSupport/Special
 RoleSupport/Special / Principal/Coordinator / Other
 RoleSupport/Special / Teaching Assistant
 RoleTeaching Assistant
 EmpStatusFreelance/Self employed
 EmpStatusOther
 EmpStatusPermanent, Full
 EmpStatusPermanent, Part
 OftenTeachIrregular/variable schedule

OftenTeachLess than once per week
OftenTeachOnce per week
OftenTeachSeveral times per week
FormalNo
FormalYes
QualifiedDoctorate
QualifiedMaster's degree
QualifiedOther
QualifiedProfessional Qualification

(Intercept)

Age
GenderMale
RoleClassroom teacher / Other
RoleClassroom teacher / Principal/Coordinator
RoleClassroom teacher / Subject Specialist
RoleClassroom teacher / Subject Specialist / Other
RoleClassroom teacher / Subject Specialist / Principal/Coordinator
RoleClassroom teacher / Subject Specialist / Support/Special
RoleClassroom teacher / Subject Specialist / Teaching Assistant
RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
RoleClassroom teacher / Support/Special
RoleClassroom teacher / Support/Special / Principal/Coordinator
RoleClassroom teacher / Teaching Assistant
RoleOther
RolePrincipal/Coordinator
RoleSubject Specialist
RoleSubject Specialist / Other
RoleSubject Specialist / Principal/Coordinator
RoleSubject Specialist / Support/Special
RoleSubject Specialist / Support/Special / Teaching Assistant
RoleSubject Specialist / Teaching Assistant / Other
RoleSupport/Special
RoleSupport/Special / Principal/Coordinator / Other
RoleSupport/Special / Teaching Assistant
RoleTeaching Assistant
EmpStatusFreelance/Self employed
EmpStatusOther
EmpStatusPermanent, Full
EmpStatusPermanent, Part
OftenTeachIrregular/variable schedule
OftenTeachLess than once per week
OftenTeachOnce per week

OftenTeachSeveral times per week
FormalNo
FormalYes
QualifiedDoctorate
QualifiedMaster's degree
QualifiedOther
QualifiedProfessional Qualification

(Intercept)

Age
GenderMale
RoleClassroom teacher / Other
RoleClassroom teacher / Principal/Coordinator
RoleClassroom teacher / Subject Specialist
RoleClassroom teacher / Subject Specialist / Other
RoleClassroom teacher / Subject Specialist / Principal/Coordinator
RoleClassroom teacher / Subject Specialist / Support/Special
RoleClassroom teacher / Subject Specialist / Teaching Assistant
RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
RoleClassroom teacher / Support/Special
RoleClassroom teacher / Support/Special / Principal/Coordinator
RoleClassroom teacher / Teaching Assistant
RoleOther
RolePrincipal/Coordinator
RoleSubject Specialist
RoleSubject Specialist / Other
RoleSubject Specialist / Principal/Coordinator
RoleSubject Specialist / Support/Special
RoleSubject Specialist / Support/Special / Teaching Assistant
RoleSubject Specialist / Teaching Assistant / Other
RoleSupport/Special
RoleSupport/Special / Principal/Coordinator / Other
RoleSupport/Special / Teaching Assistant
RoleTeaching Assistant
EmpStatusFreelance/Self employed
EmpStatusOther
EmpStatusPermanent, Full
EmpStatusPermanent, Part
OftenTeachIrregular/variable schedule
OftenTeachLess than once per week
OftenTeachOnce per week
OftenTeachSeveral times per week
FormalNo

FormalYes
 QualifiedDoctorate
 QualifiedMaster's degree
 QualifiedOther
 QualifiedProfessional Qualification

 (Intercept)
 Age
 GenderMale
 RoleClassroom teacher / Other
 RoleClassroom teacher / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist
 RoleClassroom teacher / Subject Specialist / Other
 RoleClassroom teacher / Subject Specialist / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist / Support/Special
 RoleClassroom teacher / Subject Specialist / Teaching Assistant
 RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
 RoleClassroom teacher / Support/Special
 RoleClassroom teacher / Support/Special / Principal/Coordinator
 RoleClassroom teacher / Teaching Assistant
 RoleOther
 RolePrincipal/Coordinator
 RoleSubject Specialist
 RoleSubject Specialist / Other
 RoleSubject Specialist / Principal/Coordinator
 RoleSubject Specialist / Support/Special
 RoleSubject Specialist / Support/Special / Teaching Assistant
 RoleSubject Specialist / Teaching Assistant / Other
 RoleSupport/Special
 RoleSupport/Special / Principal/Coordinator / Other
 RoleSupport/Special / Teaching Assistant
 RoleTeaching Assistant
 EmpStatusFreelance/Self employed
 EmpStatusOther
 EmpStatusPermanent, Full
 EmpStatusPermanent, Part
 OftenTeachIrregular/variable schedule
 OftenTeachLess than once per week
 OftenTeachOnce per week
 OftenTeachSeveral times per week
 FormalNo
 FormalYes
 QualifiedDoctorate

QualifiedMaster's degree
 QualifiedOther
 QualifiedProfessional Qualification

 (Intercept)
 Age
 GenderMale
 RoleClassroom teacher / Other
 RoleClassroom teacher / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist
 RoleClassroom teacher / Subject Specialist / Other
 RoleClassroom teacher / Subject Specialist / Principal/Coordinator
 RoleClassroom teacher / Subject Specialist / Support/Special
 RoleClassroom teacher / Subject Specialist / Teaching Assistant
 RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
 RoleClassroom teacher / Support/Special
 RoleClassroom teacher / Support/Special / Principal/Coordinator
 RoleClassroom teacher / Teaching Assistant
 RoleOther
 RolePrincipal/Coordinator
 RoleSubject Specialist
 RoleSubject Specialist / Other
 RoleSubject Specialist / Principal/Coordinator
 RoleSubject Specialist / Support/Special
 RoleSubject Specialist / Support/Special / Teaching Assistant
 RoleSubject Specialist / Teaching Assistant / Other
 RoleSupport/Special
 RoleSupport/Special / Principal/Coordinator / Other
 RoleSupport/Special / Teaching Assistant
 RoleTeaching Assistant
 EmpStatusFreelance/Self employed
 EmpStatusOther
 EmpStatusPermanent, Full
 EmpStatusPermanent, Part
 OftenTeachIrregular/variable schedule
 OftenTeachLess than once per week
 OftenTeachOnce per week
 OftenTeachSeveral times per week
 FormalNo
 FormalYes
 QualifiedDoctorate
 QualifiedMaster's degree
 QualifiedOther

QualifiedProfessional Qualification

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation matrix not shown by default, as $p = 40 > 12$.

Use `print(x, correlation=TRUE)` or

`vcov(x)` if you need it

optimizer (bobyqa) convergence code: 0 (OK)

boundary (singular) fit: see `help('isSingular')`

```
m_Zimet_sig <- update(m_Zimet_total, Zimet_sig_other ~ .)
```

boundary (singular) fit: see `help('isSingular')`

```
m_Zimet_fam <- update(m_Zimet_total, Zimet_family ~ .)
```

boundary (singular) fit: see `help('isSingular')`

```
m_Zimet_frd <- update(m_Zimet_total, Zimet_friends ~ .)
```

boundary (singular) fit: see `help('isSingular')`

```
# Cross-instrument model -----
m_LoiTeS_cross <- lmer(
  Michele_full ~ Tschannen_total + Zimet_total + WHO5_total +
    Age + Gender + YearTaught + EmpStatus +
    (1 | EmpCount),
  data = score_df, REML = FALSE,
  control = lmerControl(optimizer = "bobyqa")
)
```

boundary (singular) fit: see `help('isSingular')`

```
summary(m_LoiTeS_cross)
```

Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's method [lmerModLmerTest]

Formula: Michele_full ~ Tschannen_total + Zimet_total + WH05_total + Age + Gender + YearTaught + EmpStatus + (1 | EmpCount)

Data: score_df

Control: lmerControl(optimizer = "bobyqa")

AIC	BIC	logLik	-2*log(L)	df.resid
317.8	368.0	-143.9	287.8	196

Scaled residuals:

Min	1Q	Median	3Q	Max
-5.4526	-0.4323	0.0800	0.6093	2.2607

Random effects:

Groups	Name	Variance	Std.Dev.
EmpCount	(Intercept)	3.909e-16	1.977e-08
	Residual	2.290e-01	4.785e-01

Number of obs: 211, groups: EmpCount, 29

Fixed effects:

	Estimate	Std. Error	df	t value
(Intercept)	2.704899	0.426751	211.000000	6.338
Tschannen_total	0.143006	0.033878	211.000000	4.221
Zimet_total	0.164021	0.033032	211.000000	4.966
WH05_total	-0.009641	0.001991	211.000000	-4.841
Age	-0.004959	0.005556	211.000000	-0.893
GenderMale	-0.114986	0.073701	211.000000	-1.560
YearTaught>= 10 years	0.115386	0.088233	211.000000	1.308
YearTaught>= 20 years	-0.028970	0.132248	211.000000	-0.219
YearTaught>= 30 years	0.247597	0.230044	211.000000	1.076
EmpStatusFreelance/Self employed	-0.195357	0.235808	211.000000	-0.828
EmpStatusOther	0.524526	0.519459	211.000000	1.010
EmpStatusPermanent, Full	-0.015159	0.182935	211.000000	-0.083
EmpStatusPermanent, Part	-0.156450	0.194662	211.000000	-0.804

	Pr(> t)
(Intercept)	1.39e-09 ***
Tschannen_total	3.61e-05 ***
Zimet_total	1.41e-06 ***
WH05_total	2.49e-06 ***
Age	0.373
GenderMale	0.120
YearTaught>= 10 years	0.192

```

YearTaught>= 20 years          0.827
YearTaught>= 30 years          0.283
EmpStatusFreelance/Self employed 0.408
EmpStatusOther                  0.314
EmpStatusPermanent, Full        0.934
EmpStatusPermanent, Part        0.422
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Correlation matrix not shown by default, as $p = 13 > 12$.
 Use `print(x, correlation=TRUE)` or
`vcov(x)` if you need it

optimizer (bobyqa) convergence code: 0 (OK)
 boundary (singular) fit: see `help('isSingular')`

```

# Model comparison -----
m_LoiTeS_noGender <- update(m_LoiTeS_full, . ~ . - Gender)
anova(m_LoiTeS_noGender, m_LoiTeS_full)

```

Data: score_df

Models:

m_LoiTeS_noGender: Michele_full ~ Age + Role + EmpStatus + OftenTeach + Formal + Qualified +

m_LoiTeS_full: Michele_full ~ Age + Gender + Role + EmpStatus + OftenTeach + Formal + Qualif

	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
m_LoiTeS_noGender	41	437.13	574.56	-177.57	355.13			
m_LoiTeS_full	42	436.48	577.26	-176.24	352.48	2.6499	1	0.1036

```

# ICC: variance attributable to country
icc_vals <- sapply(broad_models, function(m) {
  vc      <- as.data.frame(VarCorr(m))
  country_var <- vc$vcov[vc$grp == "EmpCount"]
  resid_var  <- vc$vcov[vc$grp == "Residual"]
  country_var / (country_var + resid_var)
})
print(round(icc_vals, 3))

```

WH05_total	Michele_full	Michele_F1
0.173	0.037	0.000

Michele_F2	Michele_F3	Hughes_total
0.094	0.000	0.048
Tschannen_total	Tschannen_instruction	Tschannen_class_manager
0.000	0.136	0.000
Tschannen_engagement	Zimet_total	Zimet_sig_other
0.187	0.000	0.000
Zimet_family	Zimet_friends	
0.000	0.020	

```
# Coefficient plots (no broom.mixed - uses extract_fixed) -----
# FIX 6: plot_coefs rebuilt on extract_fixed()
plot_coefs <- function(model, title = "") {
  coef_df <- extract_fixed(model, outcome_name = "")
  coef_df %>%
    filter(term != "(Intercept)") %>%
    ggplot(aes(x = estimate, y = reorder(term, estimate))) +
    geom_vline(xintercept = 0, linetype = "dashed", colour = "grey50") +
    geom_errorbarh(aes(xmin = conf.low, xmax = conf.high),
                  height = 0.2, colour = "steelblue") +
    geom_point(size = 2.5, colour = "steelblue") +
    labs(title = title, x = "Estimate (95% CI)", y = NULL) +
    theme_bw(base_size = 11)
}

plot_coefs(m_LoiTeS_full, "LoiTeS - demographic model")
```

Warning: `geom\_errorbarh()` was deprecated in ggplot2 4.0.0.
 i Please use the `orientation` argument of `geom\_errorbar()` instead.

`height` was translated to `width`.

L1

RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
 RolePrincipal/Coordinator / Other
 EmpStatusFreelance/Self-employed
 RoleSubject Specialist / Teaching Assistant / Other
 EmpStatusPermanent, Part
 RoleClassroom teacher / Subject Specialist / Support/Special
 EmpStatusPermanent, Full
 OftenTeachOnce per week
 RoleClassroom teacher / Principal/Coordinator / Other
 RoleClassroom teacher / Principal/Coordinator / Other
 RoleSupport/Special / Principal/Coordinator / Other
 RoleSubject Specialist / Support/Special / Teaching Assistant
 FormalNo
 RoleClassroom teacher / Subject Specialist / Principal/Coordinator
 RoleSubject Specialist
 OftenTeachIrregular/Variable schedule
 RoleSubject Specialist / Principal/Coordinator
 RoleClassroom teacher / Support/Special
 FormalYes
 QualifiedMaster's degree
 RoleClassroom teacher / Subject Specialist
 OftenTeachSeveral times per week
 Age
 OftenTeachLess than once per week
 QualifiedOther
 QualifiedProfessional Qualification
 QualifiedDoctorate
 RoleClassroom teacher / Support/Special / Principal/Coordinator
 RoleOther
 RoleTeaching Assistant
 RoleSupport/Special / RoleSupport/Special
 RoleSubject Specialist / Teaching Assistant
 RoleClassroom teacher / Subject Specialist / Teaching Assistant
 RoleSubject Specialist / Other
 RoleClassroom teacher / Subject Specialist / Other
 RoleClassroom teacher / Teaching Assistant

-2

Estimate (

```
plot_coefs(m_WH05_cross, "WHO-5 - demographic model")
```

`height` was translated to `width`.

W

RoleSubject Specialist / Other
 RoleSubject Specialist / Teaching Assistant / Other
 RoleClassroom teacher / Subject Specialist / Support/Special
 QualifiedProfessional Qualification
 RoleClassroom teacher / Other
 RoleClassroom teacher / Subject Specialist / Principal/Coordinator
 QualifiedDoctorate
 OftenTeachSeveral times per week
 RoleTeaching Assistant
 RoleSupport/Special
 QualifiedMaster's degree
 RoleClassroom teacher / Subject Specialist / Teaching Assistant
 RoleClassroom teacher / Subject Specialist
 RoleOther
 Age
 QualifiedOther
 GenderMale
 RoleClassroom teacher / Support/Special
 RoleSupport/Special / Principal/Coordinator / Other
 OftenTeachIrregular/Variable schedule
 OftenTeachOnce per week
 RoleSupport/Special / Teaching Assistant
 EmpStatusOther
 RolePrincipal/Coordinator
 RoleSubject Specialist / Support/Special / RoleSubject Specialist
 RoleClassroom teacher / Principal/Coordinator
 RoleClassroom teacher / Support/Special / Principal/Coordinator
 OftenTeachLess than once per week
 RoleSubject Specialist / Support/Special
 RoleSubject Specialist / Principal/Coordinator
 EmpStatusPermanent, Full
 RoleClassroom teacher / Subject Specialist / Teaching Assistant / Principal/Coordinator / Other
 FormalNo
 RoleClassroom teacher / Teaching Assistant
 EmpStatusPermanent, Part

-40

Estimate (

```
plot_coefs(m_Tschannen_total, "Tschannen - teacher efficacy")
```

`height` was translated to `width`.

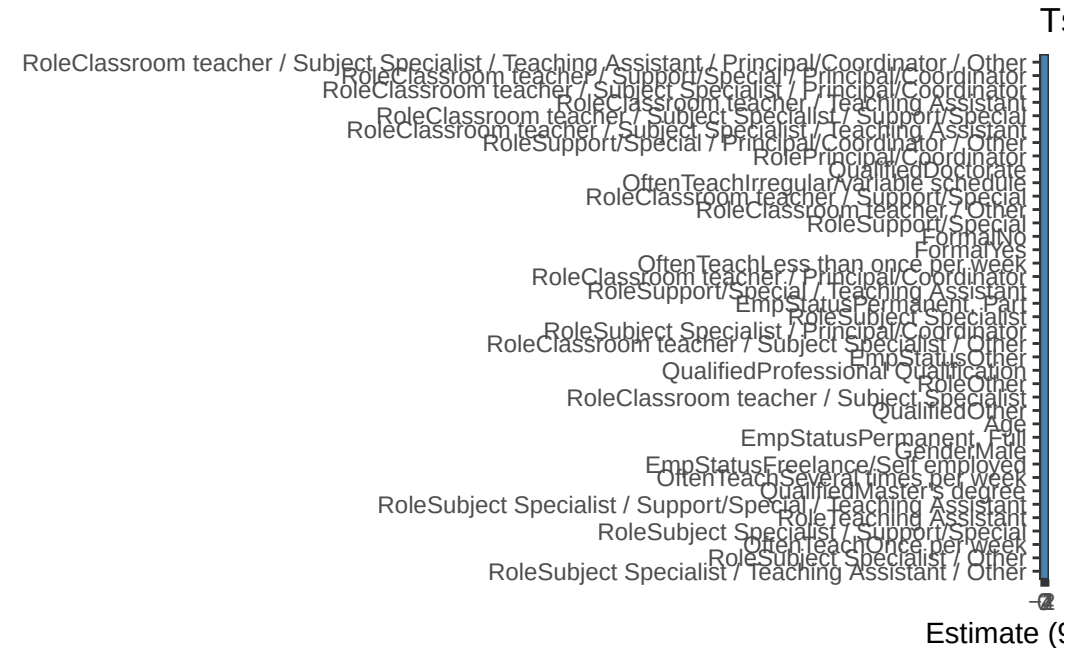


Table 1: Participant demographics (N = 211)

Variable	Level	N	%
Age	M = 39.7, SD = 8.79, Mdn = 38 [24-66]	211	
Gender	Female	144	68.2%
	Male	67	31.8%
Country	United States	66	31.3%
	United Kingdom	40	19%
	Canada	22	10.4%
	Spain	15	7.1%
	France	6	2.8%
	Australia	5	2.4%
	Germany	5	2.4%
	Brazil	4	1.9%
	Mexico	4	1.9%
	Netherlands	4	1.9%
	Poland	4	1.9%
	UAE	4	1.9%
	Argentina	3	1.4%
	Italy	3	1.4%
	Japan	3	1.4%
	Romania	3	1.4%
	South Africa	3	1.4%
	Belgium	2	0.9%
	Egypt	2	0.9%
	Indonesia	2	0.9%
	Kenya	2	0.9%
	Vietnam	2	0.9%
	India	1	0.5%
	Ireland	1	0.5%
	Morocco	1	0.5%
	Portugal	1	0.5%
	South Korea	1	0.5%
	Switzerland	1	0.5%
	Teacher	1	0.5%
Years Taught	>= 10 years	94	44.5%
	< 10 years	72	34.1%
	>= 20 years	37	17.5%
	>= 30 years	8	3.8%
Role	Classroom teacher	144	68.2%
	Subject Specialist	62	29.4%
	Support/Special	38	18%
	Teaching Assistant	20	9.5%
	Other	17	8.1%
	Principal/Coordinator	14	6.6%
Level Taught	Upper secondary	103	48.8%
	Lower secondary	102	48.3%
	Primary	91	43.1%
	Early Childhood	28	13.3%
	Further Ed./Adult	25	11.8%

Table 2: Pearson correlations between scale and sub-scale scores

	WHO-5	LoiTeS	LoiTeS F1	LoiTeS F2	LoiTeS F3	Hughes	Tschannen	Instruction	Cl. Mgmt	Engagement	Zimet	Sig. Other	Family	Friends
WHO-5														
LoiTeS	-0.50***													
LoiTeS F1	-0.32***	0.78***												
LoiTeS F2	-0.54***	0.87***	0.48***											
LoiTeS F3	-0.20**	0.60***	0.22**	0.40***										
Hughes	0.44***	-0.50***	-0.48***	-0.44***	-0.17*									
Tschannen	-0.40***	0.47***	0.19**	0.42***	0.51***	-0.30***								
Instruction	-0.30***	0.44***	0.21**	0.39***	0.48***	-0.30***	0.84***							
Cl. Mgmt	-0.31***	0.35***	0.15*	0.31***	0.37***	-0.21**	0.87***	0.58***						
Engagement	-0.44***	0.44***	0.14*	0.40***	0.49***	-0.27***	0.91***	0.67***	0.70***					
Zimet	-0.34***	0.47***	0.45***	0.37***	0.23***	-0.63***	0.36***	0.35***	0.28***	0.33***				
Sig. Other	-0.15*	0.32***	0.27***	0.26***	0.19**	-0.46***	0.26***	0.31***	0.17*	0.20**	0.82***			
Family	-0.32***	0.44***	0.45***	0.32***	0.20**	-0.49***	0.32***	0.25***	0.25***	0.32***	0.84***	0.53***		
Friends	-0.36***	0.39***	0.36***	0.32***	0.18**	-0.61***	0.32***	0.31***	0.26***	0.27***	0.79***	0.49***	0.48***	

Note. * p < .05 ** p < .01 *** p < .001 (Holm-corrected). Diagonals omitted.

Table 3: Descriptive statistics for scale and sub-scale scores

Scale	N	Mean	SD	Min	Max
WHO5_total	211	66.33	19.73	20.00	112
Michele_full	211	3.72	0.64	1.42	5
Michele_F1	211	3.63	0.90	1.00	5
Michele_F2	211	3.43	0.97	1.00	5
Michele_F3	211	4.22	0.59	2.00	5
Hughes_total	211	39.88	11.37	20.00	68
Tschannen_total	211	6.79	1.12	3.08	9
Tschannen_instruction	211	7.24	1.18	3.00	9
Tschannen_class_manager	211	6.75	1.32	2.50	9
Tschannen_engagement	211	6.40	1.33	2.75	9
Zimet_total	211	5.57	1.13	1.75	7
Zimet_sig_other	211	5.82	1.35	1.00	7
Zimet_family	211	5.37	1.51	1.00	7
Zimet_friends	211	5.52	1.29	1.00	7